TOKTICKIT / LAB 4

Final Report

Actions Taken • Ticket Workflow • Dashboards • Regression

Evidence snapshot: 1 November 2026 · final main 58b0a7a

Status: T-20 and T-25 are complete on main. The merged PR #65 tree passed 306 server tests (including 37 PostgreSQL checks), 66 client tests, and 22 browser tests; both production builds, Prisma validation, and the compiled-server health check also passed. Issues #51–#56 are Done. GitHub reported no status checks, so this report makes no CI-pass claim.

Release history: PRs #57–#65 are merged. PR #65 completed accessibility and release evidence; final-main verification was rerun after merge.

Source branch: main · merge commit 58b0a7a5556a689ca75c47f9797ebfe84c9cdb3a · tree 65731368fb2ad8c27e86da34248906a1a5881ee6

Prepared from the reviewed Lab 4 contract, closed feature Issues, verified peer reviews, post-merge local runs, and real Playwright screenshots.

Answer Part 1

Git use with engineering workflow. PRs #57–#62 merged into lab4-staging; PRs #63–#65 completed promotion, release evidence, and final-main verification. Punge089 commented and approved; book6349 replied. Issues #51–#56 are closed and all appear in Done on the project board.

PR	Scope	State at report date
#57	Contract, tests, and documentation	Merged
#58–#61	Actions Taken, UI, workflow, dashboards	Merged
#62–#65	Final regression, accessibility evidence, release checklist, and final-main verification	Merged; latest main commit 58b0a7a

PR chain: #57–#62 to lab4-staging; #63–#65 to main. Latest main merge: PR #65 at 58b0a7a5556a689ca75c47f9797ebfe84c9cdb3a.

Review record: docs/lab-04/reviewer.md

Repository evidence. Specifications and records remain under docs/lab-04, named tests under server/tests, client/tests and e2e/lab-04. Complete final-main logs are retained under output/evidence. The board was checked read-only and Issues #51–#56 are all Done.

Rendered repository and peer-review record

This renders the repository structure and the completed GitHub review record. Setup guidance: README.md; ignore rules: .gitignore; full review record: docs/lab-04/reviewer.md.

Repository layout and setup

Repository path	What the structure and setup document
docs/lab-04/	Specification, API/UI contracts, test map, AI-use, review, and screenshot index.
server/tests/lab-04/	API, role, PostgreSQL, migration, dashboard, and workflow checks.
client/tests/lab-04/	React component and interaction checks for Lab 4.
e2e/lab-04/	Real browser journeys and responsive screenshot assertions.
artifacts/lab-04/screenshots/	36 real source captures: 24 original desktop/tablet/mobile views and 12 new 320px views.
README.md / .gitignore	Stack, database setup and test commands; environment, dependencies, builds, and generated test output exclusions.

The README names Node.js 18+, Docker Desktop for PostgreSQL 18, and npm. It directs setup to a new uniquely named empty database and warns against running db push --accept-data-loss against a database containing data. The .gitignore excludes dependencies, environment files, builds, private storage, and generated Playwright output.

Actual peer-review record: PRs #57–#65

Each row links to a public GitHub PR. Punge089 is the reviewer; book6349 is the PR author. GitHub dates are UTC.

PR / date	Reviewer comment	Author reply	Reviewer approval
PR #57 · 30 Oct	Punge089: “Please confirm every acceptance criterion maps to a named test and the AI-use entries are actual prompts.” Comment	Added the acceptance-criterion map and actual prompt excerpts, with commit links. Author reply	Punge089: “The contract and test mapping are consistent, and my comments have been addressed. Approved.” Approval

PR / date	Reviewer comment	Author reply	Reviewer approval
PR #58 · 30 Oct	Punge089: “Please check actor attribution, access control, idempotency, lifecycle rules and migration recovery.” Comment	Checked those paths; reported 227 server tests (37 PostgreSQL checks), build and Prisma validation, and linked both commits. Author reply	Punge089: “Ok comments resolved, Approved.” Approval
PR #59 · 30 Oct	Punge089: “Can Requesters view Actions Taken without editing it, and does the form retain input after a failed save?” Comment	Confirmed Requesters are read-only and failed saves retain values; linked component and browser tests. Author reply	Punge089: “Requester access stays read-only, and failed saves preserve input. Looks good to merge.” Approval
PR #60 · 30 Oct	Punge089: “Try invalid status changes and blocked resolutions; neither should alter the ticket.” Comment	Confirmed rejected transitions and blocked resolution leave state unchanged; linked transition, gate, and conflict coverage. Author reply	Punge089: “Invalid transitions and blocked resolutions leave ticket state intact. I approve.” Approval
PR #61 · 31 Oct	Punge089: “Compare dashboard totals with their ticket lists, including the date-range boundaries.” Comment	Reported matching query filters, inclusive UTC windows, dashboard E2E and test/build results; Lab 1–3 and remaining Lab 4 browser checks were still pending at that time. Author reply	Punge089: “Totals match the underlying tickets across the tested date windows. Approved to merge.” Approval
PR #62 · 31 Oct	Punge089: “Please check the full-regression results and confirm the responsive captures match the tested build.”	book6349: linked the 21-test run, 24 captures, report, and commit 4f4a847.	Punge089: “The results and captures match this revision. Approved to merge.”
PR #63 · 31 Oct	Punge089: “Please confirm the staged scope and note how the pending CI and main-branch checks will be verified.”	book6349: recorded that CI and main verification were pending at the time and fixes would stay on that PR.	Punge089: “Reviewed the scope and verification notes. Approved.”
PR #64 · 31 Oct	Punge089: confirm report revision and remaining gaps.	book6349: linked commit 993af6a, verification logs and limitations.	Punge089: results and limits are clear. Approved.
PR #65 · 1 Nov	Punge089: “Please check the keyboard and 320px evidence, recovery cases, and release checklist against the recorded revision.” (comment)	book6349: “The evidence and verification limits are documented on this branch in the release notes.” (reply)	Punge089: “The accessibility evidence and release checks are clear. Approved.” (approval)

PR #65 received a reviewer comment, an author reply, and Punge089’s approval before merge. It merged to main at 58b0a7a5556a689ca75c47f9797ebfe84c9cdb3a. The reviewer identity, exact activity, and links are recorded below.

Answer Part 2

Spec DD. The Lab 4 engineering contract was reviewed in PR #57 before the feature implementation PRs. It defines numbered requirements and business rules, the Actions Taken data model, Ticket transition and resolution rules, dashboard calculations, migration and seed behavior, authorization, conflicts, safe errors, acceptance criteria, and a Product Definition of Done.

Engineering specification: specification.md

API contract: api-spec.md

UI contract: ui-spec.md

Contract PR: #57 — Contract, test DD and documentation

The acceptance criteria are traced to named tests in tests.md; the migration preserves earlier records and refuses unsafe reuse of an existing E2E database.

Rendered specification excerpts

The contract was committed in fd390c3 on 28 October 2026. PR #57 recorded Punge089’s review and approval on 30 October, before implementation PRs #58–#61 were completed. Source: specification.md.

Roles and authorization

Role	Actions Taken	Dashboards	Ticket status
Requester	Read-only; owned Tickets only	Requester dashboard; owned data only	Cannot change formal status or resolution
IT Staff	Read, create, assign, edit and transition actions on accessible Tickets	Staff dashboard and operational drill-downs	May use the documented matrix
Administrator	Same action permissions as IT Staff; may be an action assignee	Staff dashboard; no account counts	Read-only for formal status in this increment

ActionTaken record fields

Field	Contract
id	Primary key
ticketId	Required foreign key to exactly one Ticket
actionDateTime	Required effective work/plan timestamp with ISO 8601 offset; stored and compared in UTC
description	Required, trimmed, 1–2,000 characters
result	Required, trimmed, 1–2,000 characters
performedById	Required User foreign key; set from the session on create and never accepted from the client
assigneeId	Required User foreign key; active IT Staff or Administrator; distinct from Ticket Owner and performer
status	PLANNED, IN_PROGRESS, COMPLETED or CANCELLED
followUpRequired	Required boolean
followUpNote	Required, trimmed, 1–1,000 characters when followUpRequired is true; otherwise null
attachmentNotes	Optional plain text, up to 1,000 characters; describes a file to look for, not an upload
version	Integer starting at 1; incremented for each accepted edit or lifecycle change
createdAt, updatedAt	Server-managed timestamps

Ticket transition matrix

From	Allowed destinations	Confirmation required
NEW	OPEN, CANCELLED	CANCELLED
OPEN	IN_PROGRESS, WAITING_FOR_REQUESTER, RESOLVED, CANCELLED	RESOLVED, CANCELLED
IN_PROGRESS	WAITING_FOR_REQUESTER, RESOLVED, CANCELLED	RESOLVED, CANCELLED
WAITING_FOR_REQUESTER	IN_PROGRESS, RESOLVED, CANCELLED	RESOLVED, CANCELLED
RESOLVED	CLOSED, REOPENED	CLOSED
CLOSED	REOPENED	—
REOPENED	IN_PROGRESS, WAITING_FOR_REQUESTER, RESOLVED, CANCELLED	RESOLVED, CANCELLED
CANCELLED	REOPENED	—

Resolution requires confirmation, a non-empty summary, at least one COMPLETED action, and no PLANNED or IN_PROGRESS action. CANCELLED actions do not block resolution. Closing retains the summary; reopening requires a reason. The server checks the current state and workflow version in the write transaction.

All numbered functional requirements and acceptance criteria

ID	Type	Rendered contract text	Trace
FR-01	Functional requirement	Persist one or more zero-to-many ActionTaken records per Ticket; a Ticket may have no actions.	Contract

ID	Type	Rendered contract text	Trace
FR-02	Functional requirement	Validate the required fields and active Staff/Admin assignee; derive immutable performedById from the authenticated session.	Contract
FR-03	Functional requirement	Permit Requesters to list actions only for their own Tickets; deny their action writes and cross-owner access server-side.	Contract
FR-04	Functional requirement	Permit authorized Staff/Admin to create, edit and transition actions; append an event for each accepted write; never hard-delete actions.	Contract
FR-05	Functional requirement	Apply the action and Ticket status matrices, resolution gate, confirmation and version check atomically.	Contract
FR-06	Functional requirement	Return safe, consistent validation, authorization, not-found and stale-version errors without leaking another Requester's Ticket.	Contract
FR-07	Functional requirement	Provide Requester and Staff dashboards with server-calculated metrics, explicit UTC boundaries, empty states and drill-downs; Admin uses the Staff dashboard.	Contract
FR-08	Functional requirement	Migrate additively, preserve prior rows/relationships, avoid fabricated history, recover from failed rollout and seed idempotently.	Contract
FR-09	Functional requirement	Meet keyboard, focus, contrast, semantic-label, responsive and accessible error-state requirements in the UI contract.	Contract
FR-10	Functional requirement	Preserve Lab 1–3 behavior and verify server, client, browser and migration regression paths listed in the test plan.	Contract
AC-01	Acceptance criterion	Authorized Staff/Admin create an action and the server stores the authenticated performer, validated assignee, timestamp, required description/result, follow-up choice and initial version/event.	test map
AC-02	Acceptance criterion	Requesters see an ordered action history only on owned Tickets and cannot mutate it; unauthorized/cross-owner requests fail safely.	test map
AC-03	Acceptance criterion	Action edits and transitions follow the documented lifecycle, increment version and append one event; stale or duplicate requests do not partially apply.	test map
AC-04	Acceptance criterion	Every Ticket transition matches the matrix; invalid transitions and unauthorized role attempts are rejected.	test map
AC-05	Acceptance criterion	Resolution and closure enforce confirmation and the resolution gate, store server attribution/time, and preserve the resolution data in history.	test map
AC-06	Acceptance criterion	Dashboard counts use all matching rows, not a page subset; definitions, UTC boundaries, zero/empty behavior and drill-down filters match this specification.	test map
AC-07	Acceptance criterion	Migration preserves every pre-existing record and relation, creates no fabricated actions/dates, and passes rollback/recovery and twice-run seed checks.	test map
AC-08	Acceptance criterion	UI behavior is usable by keyboard and assistive technology, fits supported narrow viewports, and presents validation, authorization, conflict and loading states accessibly.	test map
AC-09	Acceptance criterion	Lab 1–3 regression suites pass with the new workflow and migration in place.	test map
AC-10	Acceptance criterion	Evidence, actual AI-use prompts, reviewer outcome, and implementation test results are recorded before the final Lab 4 release.	test map

Definition of Done requires passing named tests, resolved reviewer feedback, migration and seed recovery, role checks, dashboard/date-boundary evidence, accessible/responsive checks, and Lab 1–3 regression. The specification describes the target contract; it does not by itself prove each item is complete.

Answer Part 3

Test DD and traceability. T-20 and T-25 passed on final main at 58b0a7a5556a689ca75c47f9797ebfe84c9cdb3a: 306 server tests, 66 client tests, 22 browser tests, both builds, Prisma validation, and production health. The full logs and environment details are included in the final-main appendix. GitHub reported no status checks.

Check	Command / path	Observed result
Server regression	server/: LAB4_DB_TEST=1 npm test -- --reporter=dot	306 tests in 23 files, including 37 PostgreSQL-backed checks; exit code 0. Complete output in lab4-main-verification-validation-20261101.log.
Client regression	client/: npm test -- --reporter=dot	66 tests in 18 files; exit code 0. Complete output in lab4-main-verification-validation-20261101.log.
Browser regression	repository root: npm run test:e2e; fresh database toktickit_lab3_e2e_mainlog_20261101133650179_e47151	22 Playwright tests passed on one worker: 16 Lab 3, 2 Lab 2, 4 Lab 4; exit code 0. Complete output in lab4-main-verification-e2e-20261101.log.
Build and schema	npm --prefix server run build; npm --prefix client run build; server/: npx prisma validate	Both builds and Prisma validation passed; see lab4-main-verification-validation-20261101.log.
Fresh setup and health	Fresh E2E database, migrations, repeated seed, compiled server, GET /api/health	Migrations/seed assertions passed. Compiled node dist/src/index.js returned HTTP 200 and status=ok; see lab4-main-verification-health-20261101.log.

Test plan and AC map: docs/lab-04/tests.md

Final-main validation log: output/evidence/lab4-main-verification-validation-20261101.log

Final-main browser log: output/evidence/lab4-main-verification-e2e-20261101.log

Environment and limits. Verification ran on 1 November 2026 (UTC+07:00) at main merge commit 58b0a7a5556a689ca75c47f9797ebfe84c9cdb3a, tree 65731368fb2ad8c27e86da34248906a1a5881ee6. The tree matches PR #65 head 20b4e0a7698d3eb5527cd9eb85d36e2ce8c9daeb. PostgreSQL tests used an isolated disposable database; E2E used a fresh isolated database and one worker. No GitHub CI checks were reported. This is not a full WCAG audit, manual screen-reader evaluation, standalone axe scan, or performance benchmark.

T-20 and T-25 are complete for the named checks and recorded scope on final main. See the release record, 320px evidence, and complete final-main logs appended below.

Rendered acceptance-criterion test matrix

The full T-01–T-26 matrix follows. Statuses reflect their recorded scope; exact final-main commands, revision, review history, board state, logs, and limits are recorded in release-evidence.md.

Test	AC	Planned test and named evidence file	Status
T-01	AC-01	Server creates action; session actor immutable; active assignee validation; required fields; version 1; one creation event. Files: actions-taken.api.test.ts	Passed
T-02	AC-01, AC-03	Action field validation, trimming/length boundaries, offset timestamp normalization, follow-up conditional note and attachment note limits. Files: actions-taken.api.test.ts	Passed
T-03	AC-02	Requester sees ordered actions for owned Ticket; empty list; cross-owner and nonexistent Ticket both safe 404; Requester cannot POST/PATCH/transition. Files: actions-taken.api.test.ts	Passed
T-04	AC-03	PLANNED/IN_PROGRESS lifecycle matrix, terminal COMPLETED/CANCELLED, no hard delete, exactly one event and version increment per accepted write. Files: actions-taken.api.test.ts	Passed
T-05	AC-03	Concurrent stale edit/transition yields 409 and no partial event/data update; same idempotency key/payload returns original; different payload conflicts. Files: actions-taken.api.test.ts	Passed

Test	AC	Planned test and named evidence file	Status
T-06	AC-04	Every allowed and disallowed Ticket transition; only IT Staff can write status; administrator/requester forbidden; version conflict returns 409. Files: ticket-workflow.api.test.ts	Passed
T-07	AC-05	RESOLVED requires confirm, summary, completed action and no planned/in-progress actions; cancelled actions do not block. Files: ticket-workflow.api.test.ts	Passed
T-08	AC-05	CLOSED gate, reopen reason, server-side resolvedAt/resolvedById, retained workflow history and confirmation requirements. Files: ticket-workflow.api.test.ts	Passed
T-09	AC-06	Requester counts and drill-downs are owner scoped; active status definition; zero/empty values; not limited by list pagination. Files: requester-dashboard.api.test.ts	Passed
T-10	AC-06	Requester 7-day and 30-day exact lower/upper boundaries, future exclusion, null resolvedAt exclusion, reopened Ticket exclusion, stable top-five order. Files: requester-dashboard.api.test.ts	Passed
T-11	AC-06	Staff/Admin unassigned, assigned-to-me, per-status/priority and own-action metrics; zero values; full-count query and filter drill-down. Files: staff-dashboard.api.test.ts	Passed
T-12	AC-06	Staff recent active 7-day boundaries, future exclusion and stable top-five order; Administrator receives Staff data without user-account counts. Files: staff-dashboard.api.test.ts	Passed
T-13	AC-07	Apply additive migration to representative Lab 3 database; preserve all prior ids, relations and statuses; create no fabricated actions/resolution dates; recover from failed migration. Files: actions-taken.api.test.ts	Passed
T-14	AC-07	Seed twice with stable keys; no duplicate Users, Tickets, actions or events. Files: actions-taken.api.test.ts	Passed
T-15	AC-08	Requester read-only state, no mutation controls, ordered action history and empty state. Files: ActionsTaken.test.tsx	Passed
T-16	AC-01, AC-03, AC-08	Action form conditional fields, server validation, retained input, pending/double-submit, safe failure and conflict review state. Files: ActionsTaken.test.tsx	Passed
T-17	AC-05, AC-08	Resolution gate summary/blocker links, confirmation, advisory feedback and reopen reason UX. Files: TicketWorkflow.test.tsx	Passed
T-18	AC-06, AC-08	Requester dashboard metrics, empty/loading/error/forbidden states, failed refresh retains metrics, retry and drill-down filter recovery. client/tests/lab-04/RequesterDashboard.test.tsx	Passed
T-19	AC-06, AC-08	Staff dashboard metrics, actor/assignee labels, loading/error/forbidden states, retained metrics and filter-preserving retry. client/tests/lab-04/StaffDashboard.test.tsx	Passed
T-20	AC-08	Dedicated keyboard/focus, named regions/forms and labels, alert/aria-invalid validation, reduced motion and 320px overflow checks. e2e/lab-04/zz-accessibility.spec.ts and e2e/lab-04/evidence.ts	Passed
T-21	AC-01, AC-02, AC-03, AC-08	Staff records multiple actions; Requester sees the same history read-only; responsive widths are checked and desktop/tablet/mobile evidence is captured. Files: actions-taken-flow.spec.ts	Passed
T-22	AC-04, AC-05, AC-08	Complete Ticket resolution, close and reopen journey; blocked resolution, retained public history, and Requester write denial. Files: ticket-resolution.spec.ts	Passed
T-23	AC-06, AC-08	Requester and Staff dashboard metrics render, and active/unassigned drill-downs include the expected Tickets. Files: dashboards.spec.ts	Passed
T-24	AC-09	Final main 58b0a7a5556a689ca75c47f9797ebfe84c9cdb3a: 306 server tests (37 PostgreSQL), 66 client, 22 browser; both builds, Prisma validation and health. Complete logs in lab4-main-verification-*.log.	Passed
T-25	AC-10	Release checklist, exact final-main revision/date/commands/logs, Issues #51–#56 Done, PR #57–#65 review disposition, actual AI-use record, and stated limits. docs/lab-04/release-evidence.md	Passed
T-26	AC-09	Compiled server at final main 58b0a7a: node dist/src/index.js; GET /api/health returned HTTP 200 and status=ok. lab4-main-verification-health-20261101.log	Passed

Run identity, commands, and observed results

Check	Exact command, environment, and result
Revision / date	Final main 58b0a7a5556a689ca75c47f9797ebfe84c9cdb3a; 1 November 2026 (+07:00); tree 65731368fb2ad8c27e86da34248906a1a5881ee6, equal to tested PR #65 tree.
Server regression	306 tests in 23 files, including 37 PostgreSQL-backed checks. Command: LAB4_DB_TEST=1 npm test -- --reporter=dot in server; isolated disposable PostgreSQL test database.
Client regression	66 tests in 18 files. Command: npm test -- --reporter=dot in client.
Production builds	npm --prefix server run build and npm --prefix client run build both passed; complete output in lab4-main-verification-validation-20261101.log.
Prisma validation	npx prisma validate passed with a temporary nonconnecting DATABASE_URL because the checkout has no .env; schema validation only, no connection.
Browser regression	22 Playwright tests passed on one worker. Command: npm run test:e2e; fresh isolated database toktickit_lab3_e2e_mainlog_20261101133650179_e47151.
Fresh setup / health	Migrations and seed assertions passed. Compiled node dist/src/index.js returned HTTP 200 and status=ok; see lab4-main-verification-health-20261101.log.
Release boundary	Final-main results; PR #65 is merged. Issues #51–#56 are verified Done. GitHub reported no status checks; no CI pass is claimed.

Historical candidate verification logs (31 October)

The following full validation and browser logs were produced on candidate 3ddd393 before PR #65 merged. They are preserved as historical evidence; the final-main runs at 58b0a7a are appended at the end of this report.

Historical candidate server, client, build, and Prisma output

Historical candidate revision: 3ddd3930b2c83d3193164257d3bf7dd3bcfb14e9; 2026-10-31 Asia/Bangkok

```
> toktickit-server@1.0.0 test
> vitest run --reporter=dot


 RUN  v2.1.9 C:/Users/Jitvoottikrai/OneDrive/เดสก์ท็อป/SoftwareEn/LAB1/Lab1_Starter_Scaffold/toktickit/LAB4/toktickit-main-verify-6e66c2c/server

 ✓ tests/lab-03/migration-regression.test.ts (2 tests) 21ms
 ✓ tests/lab-03/auth.unit.test.ts (4 tests) 419ms
 ✓ tests/lab-02/validation.unit.test.ts (3 tests) 5ms
 ✓ tests/lab-03/e2e-fixtures.unit.test.ts (11 tests) 763ms
   ✓ Lab 3 disposable E2E preparation safeguards > uses actual email-based IDs, preserves accounts, and populates stable realistic Queue fixtures 517ms
 ✓ tests/lab-04/staff-dashboard.api.test.ts (3 tests) 126ms
 ✓ tests/lab-02/my-tickets.api.test.ts (2 tests) 100ms
 ✓ tests/lab-02/requester-context.api.test.ts (2 tests) 123ms
 ✓ tests/lab-02/ticket-detail.api.test.ts (2 tests) 136ms
 ✓ tests/lab-04/requester-dashboard.api.test.ts (4 tests) 207ms
 ✓ tests/lab-02/create-ticket.api.test.ts (2 tests) 242ms
 ✓ tests/lab-02/attachments.api.test.ts (3 tests) 287ms
 ✓ tests/lab-03/authorization.api.test.ts (3 tests) 367ms
 ✓ tests/lab-02/ticket-number.unit.test.ts (1 test) 22ms
 ✓ tests/lab-03/comments-notes.api.test.ts (8 tests) 685ms
 ✓ tests/lab-03/staff-queue.api.test.ts (38 tests) 751ms
 ✓ tests/lab-02/reference-data.api.test.ts (2 tests) 90ms
 ✓ tests/lab-04/ticket-workflow.api.test.ts (72 tests) 1713ms
   ✓ Lab 4 Ticket workflow API > enforces 'NEW' -> 'NEW' (allowed=false) 366ms
 ✓ tests/lab-01/categories.test.ts (1 test) 48ms
 ✓ tests/lab-03/auth.api.test.ts (7 tests) 1789ms
   ✓ Lab 3 authentication contract > requires CSRF and returns safe user data on login 589ms
   ✓ Lab 3 authentication contract > gates normal routes until password change and revokes the old session 679ms
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts (89 tests) 1827ms
   ✓ Lab 3 IT Staff Ticket Detail > requires the approved confirmation and transition matrix on the backend 301ms
```

```
 ✓ tests/lab-01/health.test.ts (1 test) 51ms
 ✓ tests/lab-03/users-admin.api.test.ts (9 tests) 3715ms
   ✓ Lab 3 Administrator User Management API > lists safe users with case-insensitive search and one role filter 506ms
   ✓ Lab 3 Administrator User Management API > rejects non-Administrators and invalid role filters 755ms
   ✓ Lab 3 Administrator User Management API > creates an active user with a hashed initial password and forced change 423ms
   ✓ Lab 3 Administrator User Management API > rejects duplicate email and invalid create input safely 623ms
   ✓ Lab 3 Administrator User Management API > edits permitted account fields and resets the initial password 360ms
   ✓ Lab 3 Administrator User Management API > blocks self-deactivation and removal of the last active Administrator 401ms
stdout | tests/lab-04/actions-taken.api.test.ts > Lab 4 Actions Taken with disposable PostgreSQL
Disposable database: toktickit_lab4_prep_1793463020274

 ✓ tests/lab-04/actions-taken.api.test.ts (37 tests) 17849ms
   ✓ Lab 4 Actions Taken with disposable PostgreSQL > rolls back action, event and Ticket version when persistence fails 925ms

 Test Files  23 passed (23)
      Tests  306 passed (306)
   Start at  23:10:15
   Duration  22.75s (transform 2.48s, setup 0ms, collect 54.76s, tests 31.34s, environment 6ms, prepare 20.81s)


> toktickit-client@1.0.0 test
> vitest run --reporter=dot


 RUN  v2.1.9 C:/Users/Jitvoottikrai/OneDrive/เดสก์ท็อป/SoftwareEn/LAB1/Lab1_Starter_Scaffold/toktickit/LAB4/toktickit-main-verify-6e66c2c/
client

 ✓ tests/lab-01/App.test.tsx (3 tests) 470ms
   ✓ App > renders the TokTickIT sign-in heading after checking the session 420ms
 ✓ tests/lab-03/ui-style.test.tsx (2 tests) 597ms
   ✓ Lab 3 Zen Green styling and responsive rules > renders accessible sign-in controls with the primary action hierarchy 590ms
 ✓ tests/lab-02/RequesterContext.test.tsx (1 test) 741ms
   ✓ Lab 2 requester context > restores the authenticated Requester and ignores stale development-selector state 724ms
 ✓ tests/lab-03/ChangePassword.test.tsx (2 tests) 1475ms
   ✓ Lab 3 mandatory Change Password > blocks the normal shell until a valid replacement password is saved 1301ms
 ✓ tests/lab-03/Login.test.tsx (3 tests) 1685ms
   ✓ Lab 3 Login > shows the login form and enters the authenticated Requester shell without a selector 1089ms
   ✓ Lab 3 Login > shows and disables the visible sign-in busy state while authentication is pending 301ms
 ✓ tests/lab-02/ui-style.test.tsx (2 tests) 1703ms
   ✓ Lab 2 Zen Green UI conventions > exposes the required theme tokens and visible button hierarchy 1238ms
   ✓ Lab 2 Zen Green UI conventions > keeps active navigation and focusable labeled controls visible 440ms
 ✓ tests/lab-02/RequesterTicketDetail.test.tsx (2 tests) 1902ms
   ✓ Lab 2 owned Ticket Detail > renders an owned detail as read-only and navigates back to My Tickets 1580ms
   ✓ Lab 2 owned Ticket Detail > shows a safe error when the detail is not owned or no longer exists 319ms
 ✓ tests/lab-03/UserManagement.test.tsx (3 tests) 2013ms
   ✓ Lab 3 Administrator User Management > lists users, applies search and role filters, and keeps the direct detail entry point 751ms
   ✓ Lab 3 Administrator User Management > creates a user with clear initial-password behavior 585ms
   ✓ Lab 3 Administrator User Management > edits an account and exposes reset initial password as a separate action 666ms
 ✓ tests/lab-03/StaffTicketQueue.test.tsx (2 tests) 1945ms
   ✓ Lab 3 IT Staff Ticket Queue > loads the authenticated Staff Queue and sends the documented filters 1436ms
   ✓ Lab 3 IT Staff Ticket Queue > distinguishes an empty Queue from a filtered no-result state 471ms
 ✓ tests/lab-02/AttachmentSection.test.tsx (1 test) 1915ms
   ✓ Lab 2 AttachmentSection > allows upload and removal for active files while hiding actions for removed files 1882ms
 ✓ tests/lab-02/MyTickets.test.tsx (2 tests) 1986ms
   ✓ Lab 2 My Tickets > sends the selected filters and sort controls to the requester-scoped API 1354ms
   ✓ Lab 2 My Tickets > supports pagination and distinguishes a filtered no-results state 630ms
 ✓ tests/lab-04/RequesterDashboard.test.tsx (6 tests) 2293ms
   ✓ Lab 4 Requester dashboard UI > shows UTC context, full counts, accessible definitions and matching owned-Ticket drill-downs 662ms
   ✓ Lab 4 Requester dashboard UI > retains visible zero metrics and lets the user return from an empty drill-down 322ms
   ✓ Lab 4 Requester dashboard UI > announces initial failure Network request failed. and recovers through Retry 429ms
   ✓ Lab 4 Requester dashboard UI > announces initial loading and keeps the last metrics after a failed refresh 365ms
   ✓ Lab 4 Requester dashboard UI > announces drill-down loading and retries the same filter after failure 321ms
 ✓ tests/lab-04/StaffDashboard.test.tsx (6 tests) 2345ms
   ✓ Lab 4 Staff dashboard UI > shows all status and IT-priority values and drills down through the dashboard filter API 899ms
   ✓ Lab 4 Staff dashboard UI > keeps performer and assignee distinct and opens the linked Ticket detail 401ms
   ✓ Lab 4 Staff dashboard UI > announces initial loading and keeps the last metrics after a failed refresh 326ms
 ✓ tests/lab-03/RequesterWorkflow.test.tsx (3 tests) 2588ms
   ✓ Lab 3 Requester ticket regression > adds a public comment to the owned ticket and never renders Internal Notes 1727ms
   ✓ Lab 3 Requester ticket regression > shows the Requester resolution indication after the API confirms it 554ms
   ✓ Lab 3 Requester ticket regression > renders comment markup as text without creating executable HTML 304ms
 ✓ tests/lab-02/CreateTicket.test.tsx (4 tests) 2469ms
   ✓ Lab 2 Create Ticket > shows field-level validation without calling the API 1217ms
   ✓ Lab 2 Create Ticket > rejects an invalid attachment in the browser 333ms
   ✓ Lab 2 Create Ticket > disables duplicate submission and shows the backend ticket number on success 524ms
```

```
  ✓ Lab 2 Create Ticket > keeps entered values after an API failure 386ms
✓ tests/lab-04/ActionsTaken.test.tsx (11 tests) 3147ms
  ✓ Lab 4 Actions Taken UI > shows multiple actions in order with performer and assignee clearly separated for a Requester 461ms
  ✓ Lab 4 Actions Taken UI > announces assignee loading and keeps Save disabled until valid choices arrive 566ms
  ✓ Lab 4 Actions Taken UI > lets IT_STAFF create distinct actions with an offset, conditional follow-up and idempotency key 425ms
  ✓ Lab 4 Actions Taken UI > lets ADMINISTRATOR create distinct actions with an offset, conditional follow-up and idempotency key 373ms
  ✓ Lab 4 Actions Taken UI > keeps form values and focuses invalid input after a server validation error 307ms
✓ tests/lab-03/StaffTicketDetail.test.tsx (6 tests) 3222ms
  ✓ Lab 3 Staff Ticket Detail > shows a Requester resolution indication without changing formal status for 'IT_STAFF' 1151ms
  ✓ Lab 3 Staff Ticket Detail > shows a Requester resolution indication without changing formal status for 'ADMINISTRATOR' 315ms
  ✓ Lab 3 Staff Ticket Detail > does not show an indication when none exists for 'IT_STAFF' 462ms
  ✓ Lab 3 Staff Ticket Detail > keeps public comments and Internal Notes separate and supports staff operations 972ms
✓ tests/lab-04/TicketWorkflow.test.tsx (7 tests) 3287ms
  ✓ Lab 4 Ticket workflow UI > uses only the permitted transitions returned by the server 831ms
  ✓ Lab 4 Ticket workflow UI > shows resolution blockers and links to Actions Taken 804ms
  ✓ Lab 4 Ticket workflow UI > sends the current version, confirmation, and resolution summary 367ms
  ✓ Lab 4 Ticket workflow UI > requires a reopening reason before enabling the status change 413ms
  ✓ Lab 4 Ticket workflow UI > refreshes Ticket details and the resolution gate after an Actions Taken change without clearing the form 477ms

 Test Files  18 passed (18)
      Tests  66 passed (66)
   Start at  23:10:41
   Duration  17.98s (transform 3.52s, setup 26.04s, collect 35.50s, tests 35.78s, environment 160.04s, prepare 20.33s)


> toktickit-server@1.0.0 build
> tsc


> toktickit-client@1.0.0 build
> tsc && vite build

vite v6.4.3 building for production...
transforming...
✓ 33 modules transformed.
rendering chunks...
computing gzip size...
dist/index.html                  0.41 kB │ gzip:  0.28 kB
dist/assets/index-CgOHpO3y.css  246.77 kB │ gzip: 34.38 kB
dist/assets/index-Bq3yUVKn.js   238.44 kB │ gzip: 66.41 kB
✓ built in 2.12s
Prisma schema loaded from prisma\schema.prisma
The schema at prisma\schema.prisma is valid 🚀
Validation exit code: 0
```

Complete browser regression output

```
Revision: 3ddd3930b2c83d3193164257d3bf7dd3bcfb14e9; 2026-10-31 Asia/Bangkok

> toktickit-lab3-integration@1.0.0 test:e2e
> npm run test:e2e:prepare && playwright test


> toktickit-lab3-integration@1.0.0 test:e2e:prepare
> npm --prefix server run test:e2e:prepare


> toktickit-server@1.0.0 test:e2e:prepare
> tsx prisma/prepare-lab3-e2e.ts

New disposable database: toktickit_lab3_e2e_release_complete_20261031231014
server: npx --no-install prisma db execute --schema prisma/schema.prisma --stdin
Script executed successfully.
Synthetic pre-Lab-3 snapshot: {"requesters":[{"id":1,"name":"Ariya
Somchai","email":"ariya.somchai@example.com","isActive":true,"createdAt":"2026-10-31T16:10:19.860Z"}],"tickets":[{"id":1,"ticketNumber":"TT-
DOCKER-PRELAB3-001","requesterId":1,"categoryId":1,"relatedSystemId":1,"requestedPriority":"HIGH","status":"NEW","summary":"Pre-Lab 3
preservation fixture","description":"Synthetic migration fixture for local verification."}],"attachments":
[{"id":1,"ticketId":1,"originalFilename":"baseline.txt","storageKey":"toktickit_lab3_e2e_release_complete_20261031231014-
baseline.txt","mimeType":"text/plain","sizeBytes":17,"removedAt":null}],"categories":[{"id":1,"name":"Hardware","isActive":true,"createdAt":"2026-
10-31T16:10:19.860Z"}],"systems":[{"id":1,"name":"Corporate Laptop","isActive":true,"createdAt":"2026-10-31T16:10:19.860Z"}]}
server: npx --no-install prisma migrate resolve --applied 20260824000000_lab2_foundation
Prisma schema loaded from prisma\schema.prisma
Datasource "db": PostgreSQL database "toktickit_lab3_e2e_release_complete_20261031231014", schema "public" at "localhost:5432"

Migration 20260824000000_lab2_foundation marked as applied.
```

```
server: npx --no-install prisma migrate deploy
Prisma schema loaded from prisma\schema.prisma
Datasource "db": PostgreSQL database "toktickit_lab3_e2e_release_complete_20261031231014", schema "public" at "localhost:5432"

4 migrations found in prisma/migrations

Applying migration `20260919000000_lab3_auth_requester`
Applying migration `20261030000000_lab4_actions_foundation`
Applying migration `20261030100000_lab4_ticket_workflow`

The following migration(s) have been applied:

migrations/
  └─ 20260919000000_lab3_auth_requester/
    └─ migration.sql
  └─ 20261030000000_lab4_actions_foundation/
    └─ migration.sql
  └─ 20261030100000_lab4_ticket_workflow/
    └─ migration.sql

All migrations have been successfully applied.
server: npx --no-install prisma migrate status
Prisma schema loaded from prisma\schema.prisma
Datasource "db": PostgreSQL database "toktickit_lab3_e2e_release_complete_20261031231014", schema "public" at "localhost:5432"

4 migrations found in prisma/migrations

Database schema is up to date!
Post-migration snapshot: {"requesters":[{"id":1,"name":"Ariya Somchai","email":"ariya.somchai@example.com","isActive":true,"createdAt":"2026-
10-31T16:10:19.860Z"}],"tickets":[{"id":1,"ticketNumber":"TT-DOCKER-PRELAB3-
001","requesterId":1,"categoryId":1,"relatedSystemId":1,"requestedPriority":"HIGH","status":"NEW","summary":"Pre-Lab 3 preservation
fixture","description":"Synthetic migration fixture for local verification."}],"attachments":
[{"id":1,"ticketId":1,"originalFilename":"baseline.txt","storageKey":"toktickit_lab3_e2e_release_complete_20261031231014-
baseline.txt","mimeType":"text/plain","sizeBytes":17,"removedAt":null}],"categories":[{"id":1,"name":"Hardware","isActive":true,"createdAt":"2026-
10-31T16:10:19.860Z"}],"systems":[{"id":1,"name":"Corporate Laptop","isActive":true,"createdAt":"2026-10-31T16:10:19.860Z"}]}
server: npm run prisma:seed (run 1)

> toktickit-server@1.0.0 prisma:seed
> tsx prisma/seed.ts

Seeding completed successfully: 10 users and 3 tickets.
Seed 1 assertions:
{"users":10,"tickets":4,"attachments":1,"comments":3,"notes":3,"activeRequesters":4,"inactiveRequesters":1,"activeStaff":3,"inactiveStaff":1,"activ
eAdmins":1,"seededRequesterOwnership":3,"seededStaffOwnership":2,"commentAuthors":3,"noteAuthors":3,"baselineAttachment":1}
server: npm run prisma:seed (run 2)

> toktickit-server@1.0.0 prisma:seed
> tsx prisma/seed.ts

Seeding completed successfully: 10 users and 3 tickets.
Seed 2 assertions:
{"users":10,"tickets":4,"attachments":1,"comments":3,"notes":3,"activeRequesters":4,"inactiveRequesters":1,"activeStaff":3,"inactiveStaff":1,"activ
eAdmins":1,"seededRequesterOwnership":3,"seededStaffOwnership":2,"commentAuthors":3,"noteAuthors":3,"baselineAttachment":1}
Baseline attachment SHA256: cc7098632c30d31ed81ff5936a75a027f3136024cb96efd35e8fe3f9b4d3cfaf
Prepared E2E fixture IDs/count:
{"databaseName":"toktickit_lab3_e2e_release_complete_20261031231014","baselineTicketId":1,"reassignmentOwnerId":8,"queueTotal":20}
[WebServer] (node:33488) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (Use `node --trace-warnings ...` to show where the warning was created)
[WebServer] (node:17528) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (Use `node --trace-warnings ...` to show where the warning was created)
[WebServer] (node:6408) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (Use `node --trace-warnings ...` to show where the warning was created)

Running 22 tests using 1 worker

(node:11740) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
(Use `node --trace-warnings ...` to show where the warning was created)
  ok  1 [lab-03] › e2e\lab-03\authentication.spec.ts:15:3 › Lab 3 authentication and Requester regression › valid login, first-login change, role
shell, logout, and direct blocking (2.5s)
  ok  2 [lab-03] › e2e\lab-03\authentication.spec.ts:33:3 › Lab 3 authentication and Requester regression › invalid and inactive accounts receive
safe sign-in failure (2.1s)
  ok  3 [lab-03] › e2e\lab-03\authentication.spec.ts:50:3 › Lab 3 authentication and Requester regression › Requester can create, open,
comment, and indicate resolution (3.4s)
```

```
  ok  4 [lab-03] › e2e\lab-03\authentication.spec.ts:73:3 › Lab 3 authentication and Requester regression › Requester shell is captured at required responsive sizes (2.4s)
  ok  5 [lab-03] › e2e\lab-03\staff-ticket-flow.spec.ts:5:3 › Lab 3 IT Staff Queue and Ticket Detail › IT Staff can search Queue, open Detail, and see separate comments and notes (3.1s)
  ok  6 [lab-03] › e2e\lab-03\staff-ticket-flow.spec.ts:23:3 › Lab 3 IT Staff Queue and Ticket Detail › IT Staff Queue and Detail are captured at required responsive sizes (3.3s)
  ok  7 [lab-03] › e2e\lab-03\staff-ticket-flow.spec.ts:30:3 › Lab 3 IT Staff Queue and Ticket Detail › IT Staff can advance to a populated second Queue page (3.1s)
  ok  8 [lab-03] › e2e\lab-03\staff-ticket-flow.spec.ts:46:3 › Lab 3 IT Staff Queue and Ticket Detail › preserves migrated Attachment and verifies reassignment, priority, and status operations (2.5s)
  ok  9 [lab-03] › e2e\lab-03\staff-ticket-flow.spec.ts:75:3 › Lab 3 IT Staff Queue and Ticket Detail › Staff sees the Requester resolution indication without a formal status change (2.0s)
  ok 10 [lab-03] › e2e\lab-03\user-administration.spec.ts:5:3 › Lab 3 Administrator User Management › Administrator can list, search, create, edit, reset, and require the new initial password at next login (3.9s)
  ok 11 [lab-03] › e2e\lab-03\user-administration.spec.ts:60:3 › Lab 3 Administrator User Management › Administrator navigation excludes Queue and direct Ticket Detail remains explicit (1.1s)
  ok 12 [lab-03] › e2e\lab-03\user-administration.spec.ts:70:3 › Lab 3 Administrator User Management › Administrator User Management is captured at required responsive sizes (2.4s)
  ok 13 [lab-03] › e2e\lab-03\z-accessibility-evidence.spec.ts:24:3 › Lab 3 accessibility and transient-state evidence › captures the visible sign-in busy state and successful destination (2.5s)
  ok 14 [lab-03] › e2e\lab-03\z-accessibility-evidence.spec.ts:40:3 › Lab 3 accessibility and transient-state evidence › completes a keyboard-only sign-in and reaches requester navigation (1.6s)
  ok 15 [lab-03] › e2e\lab-03\z-accessibility-evidence.spec.ts:57:3 › Lab 3 accessibility and transient-state evidence › traverses the IT Staff Queue and opens Ticket Detail with the keyboard (2.1s)
  ok 16 [lab-03] › e2e\lab-03\z-accessibility-evidence.spec.ts:69:3 › Lab 3 accessibility and transient-state evidence › traverses Administrator User Management and opens Create User with the keyboard (1.5s)
(node:17316) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
(Use `node --trace-warnings ...` to show where the warning was created)
  ok 17 [lab-02] › e2e\lab-02\requester-ticket-flow.spec.ts:24:3 › Lab 2 requester ticket flow › creates, lists, inspects, and manages an owned ticket (4.2s)
  ok 18 [lab-02] › e2e\lab-02\requester-ticket-flow.spec.ts:79:3 › Lab 2 requester ticket flow › supports keyboard traversal across authenticated requester controls (1.4s)
(node:13252) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
(Use `node --trace-warnings ...` to show where the warning was created)
  ok 19 [lab-04] › e2e\lab-04\actions-taken-flow.spec.ts:27:3 › Lab 4 Actions Taken Ticket Detail › Staff records two actions on one Ticket and its Requester sees both read-only (3.7s)
  ok 20 [lab-04] › e2e\lab-04\dashboards.spec.ts:6:3 › Lab 4 role dashboards › shows Requester and Staff metrics and opens the matching paginated Ticket lists (2.9s)
  ok 21 [lab-04] › e2e\lab-04\ticket-resolution.spec.ts:6:3 › Lab 4 Ticket resolution workflow › resolves, closes, and reopens a Ticket with a Requester-readable history (4.3s)
  ok 22 [lab-04] › e2e\lab-04\zz-accessibility.spec.ts:25:1 › Lab 4 keyboard focus, labels, validation, reduced motion and 320px layouts (3.2s)

  22 passed (1.2m)
E2E exit code: 0
```

Compiled server health output

```
Revision: 3ddd3930b2c83d3193164257d3bf7dd3bcfb14e9; 2026-10-31 Asia/Bangkok
Command: node dist/src/index.js; GET /api/health; HTTP 200
{"status":"ok","service":"TokTickIT API"}
```

Answer Part 4

AI use with reflection. The committed AI-use record names the model and includes actual prompt excerpts used for the Lab 4 specification and implementation. It is the source for the selected prompts and records what was checked against code and test results.

AI-use record: docs/lab-04/ai-use.md

My Reflection. The specification agent helped turn the stakeholder request into numbered rules and acceptance criteria before implementation. The coding agent was most useful when work stayed tied to an Issue, a branch, and named tests. I learned to verify suggestions against the running application and database: a build alone did not catch integration problems such as stale selectors or role-flow timing. Keeping test output and screenshots tied to a commit made review more concrete. I still had to judge what was safe to claim, especially when the final staging PR and release review were not complete.

Actual prompt excerpts rendered from the AI-use record

Model: GPT-6 in Codex; the exact served deployment identifier is not exposed. Source: docs/lab-04/ai-use.md.

1. read handover file and plans thing out for SELAB4 tasks

2. did u planout pr/issues?

3. For Lab 4, we can create all the **approved Issues in one session**, as separate Issues, and add each to the project board in Backlog. Then work through them one at a time: only the Issue you're actively implementing goes to Started, and only one PR should be active at a time. We still need to agree on the Lab 4 issue list before creating them.

4. while we wating for my friend to come, preprare content for the next PRs

5. For each PR, my friend reviews and leaves comments. I reply to every comment and keep any fixes on the same branch and PR. My friend—not the PR author—approves and merges. Please prepare short, copy-ready comment, reply, and approval text, but don't post or approve on my friend's behalf.

6. can u also prepare content for the next pr planned? my friebd is out side and will be back in a few hour, when hes back we can go faster after closing this pr, also these comment/reply/approve only for pr not issue

Reflection from the source record: AI assistance helped organize the approved handout into explicit requirements and connect acceptance criteria to planned tests. This PR contains documentation work only; no application code was changed and no planned tests were run for it. Later coding preparation was made separately for the next Issue. Peer review and merge remain human decisions.

Answer Part 5

Working IT Staff Dashboard. The Staff/Admin dashboard reports unassigned work, Tickets owned by the current staff member, status and priority counts, recent active Tickets, and the current user's Actions Taken. Metrics are calculated on the server from authoritative Ticket data and link to filtered lists. The API tests compare each count to fixture records and verify the matching drill-down filters; the browser flow opens the corresponding paginated lists.

Server metric tests: staff-dashboard.api.test.ts

UI tests: StaffDashboard.test.tsx

Browser flow: dashboards.spec.ts

The responsive dashboard screenshots are included in the evidence pages at the end of this report. Zero/empty states and role-specific access are covered by the client and API checks.

Dashboard metric-to-query trace

Surface	Database / named test evidence	What it establishes
Requester counts / recent lists	server/src/dashboards.ts scopes by authenticated requesterId; Prisma count supplies all matches and findMany limits recent rows to five with inclusive UTC bounds. query · API test · UI test	Asserts owner scope, full counts, 7/30-day windows, matching drill-downs, zero metrics, and the empty-list return path.
Staff / Admin metrics	Server counts active, unassigned, current-user, assigned-action and recent scopes; status and priority totals use Prisma groupBy. query · API test · UI test	Covers zero groups, Administrator access, Requester denial, drill-down predicates, and separate performer/assignee labels.
Database drill-down	The detail route applies the same active-status, owner, priority, and date predicates to count and paginated rows. query · API test · browser journey	Confirms counts do not come from the current page and that role dashboards open matching Ticket lists.
Loading / failed UI requests	RequesterDashboard.test.tsx and StaffDashboard.test.tsx test initial loading, network/forbidden errors, retry, retained metrics after failed refresh, drill-down loading and filter-preserving recovery.	Eight added cases passed in the final 66-test client run. Error and status roles are asserted; stale list rows are not shown during failure.

Answer Part 6

Working Actions Taken. Each Ticket can hold multiple chronological Actions Taken entries with action time, description, result, server-attributed performer, follow-up flag and conditional note, and attachment notes. The primary Ticket Owner remains distinct from the staff member who performs an action. IT Staff and Administrators can write on accessible Tickets; Requesters can read their own Ticket history but cannot edit it.

Actions Taken API and PostgreSQL tests: actions-taken.api.test.ts

Actions Taken UI tests: ActionsTaken.test.tsx

Actions Taken browser flow: actions-taken-flow.spec.ts

Verification. The 37 PostgreSQL-backed checks cover persistence, actor and assignee validation, lifecycle rules, optimistic conflicts, idempotency, and migration recovery. The browser test records two actions on one Ticket and verifies the Requester sees both as read-only. The desktop, tablet, and mobile form/history captures follow.

Behavior evidence not visible in the static captures

The screenshots show the entry form, saved multi-action history, and Requester read-only history. They do not show the interactions below; each claim links to a named component/API test.

Behavior	Screenshot coverage	Named test evidence
Edit an in-progress action	Not shown.	UI edit/version · API stale/cross-Ticket guard
Complete / cancel	Not shown.	UI terminal confirmation · API lifecycle matrix · terminal immutability
Validation / retained input	T-20 shows validation alert and invalid-field focus; retained input is covered by tests.	UI validation and focus · API input validation
Inactive-assignee rejection	Not shown.	API active eligible assignee
Requester role / multi-action Ticket	Read-only history and saved entries are shown; server role checks also apply.	Requester API scope · component history · same-Ticket browser journey
Safe failure / duplicate / concurrency	Not shown.	UI safe failure · concurrent update · duplicate create · rollback

The Actions Taken browser journey verifies two recorded actions and Requester read-only history. The dedicated T-20 journey also checks validation alerts and focus. Editing, terminal actions and inactive-assignee rejection rely on the linked component and API tests.

Answer Part 7

Working Ticket Workflow. The server enforces the documented transition matrix and resolution gate. A Requester's "problem appears resolved" response is advisory. Formal resolution requires the contract's confirmation and action-completion rules; invalid transitions, unresolved blockers, stale versions, and forbidden roles are rejected without partial updates. Reopen events retain history.

Workflow API tests: ticket-workflow.api.test.ts

Workflow UI tests: TicketWorkflow.test.tsx

Resolution browser flow: ticket-resolution.spec.ts

The 72 workflow API tests and final browser journey passed. Evidence pages show a blocked resolution, a reopened Ticket, and the Requester-visible resolution history.

Ticket workflow transition and history evidence

Required behavior	Direct source links
Allowed / rejected transitions	API transition matrix (72 cases) · UI legal choices
Resolution gate and cancellation rule	API completed-action gate · UI blocker link
Confirmation, close, retained summary	API confirmation/history
Reopen reason and history	API reopen test · browser resolution journey
Stale version and atomic update	Ticket workflow API conflict · Action stale update

Required behavior	Direct source links
Role visibility and owner scope	Requester/Admin read-only status · owned history · cross-owner safe 404

Answer Part 8

Working Requester Dashboard and Final Regression UI. Owner-scoped counts and matching drill-downs are verified by API, component and browser tests. Final main passed 306 server tests, 66 client tests and 22 browser tests. Dashboard loading, safe network/forbidden errors, retry, retained metrics after failed refresh and filter-preserving drill-down retry are covered for both roles.

Requester metric tests: requester-dashboard.api.test.ts

Requester dashboard UI tests: RequesterDashboard.test.tsx

Regression configuration: playwright.config.ts

Scope note. T-20 and T-25 passed on main commit 58b0a7a5556a689ca75c47f9797ebfe84c9cdb3a on 1 November 2026 (+07:00). The merged tree passed 306 server tests, 66 client tests, 22 browser tests, both builds, Prisma validation and production health. Issues #51–#56 are verified in Done. GitHub reported no status checks.

Regression coverage verified on final main

Regression layer	Observed coverage in the committed run logs
Lab 1	Health and category suites passed in the server/client regression logs.
Lab 2	Requester create/list/detail, filters, keyboard traversal, attachments, and shared UI conventions appear in named client/browser output.
Lab 3	Authentication and roles, Staff Queue/Detail, Requester public comments vs Internal Notes, Admin User Management, migration-fixture preservation, and keyboard journeys appear in the run logs.
Lab 4	Actions Taken multi-entry/read-only flow, both dashboards with matching drill-downs, and Ticket resolve/close/reopen browser flow passed.

The final-main E2E run passed 22 tests: 16 Lab 3, 2 Lab 2 and 4 Lab 4 journeys, using one worker and a new isolated PostgreSQL database. Complete output is in the final-main appendix and evidence log.

Answer Part 9

Zen Green UI Responsive Behavior Accessibility and Final Polish. The report retains 24 original desktop/tablet/mobile captures. Twelve new 320px captures cover all eight existing Lab 4 states plus four dedicated keyboard/validation states. The dedicated T-20 journey checks keyboard reachability, visible focus, named regions/forms, associated labels, validation alerts and aria-invalid focus, reduced motion, and page overflow at 320px. The original captures predate the focus-outline correction; new captures show the current implementation.

Screenshot index: docs/lab-04/screenshots.md

Review item	Evidence	Status
Responsive layouts	Dedicated T-20 keyboard/focus/labels/validation/reduced-motion journey and eight screen-state overflow assertions at 320px; full 22-test browser run passed.	Passed
Keyboard and visible focus	Dedicated T-20 keyboard/focus/labels/validation/reduced-motion journey and eight screen-state overflow assertions at 320px; full 22-test browser run passed.	Passed
Labels, live regions, reduced motion, non-color status cues	Dedicated T-20 keyboard/focus/labels/validation/reduced-motion journey and eight screen-state overflow assertions at 320px; full 22-test browser run passed.	Passed

Review item	Evidence	Status
Standalone automated accessibility scan	No separate axe or equivalent audit was run	Not run
GitHub CI and main-branch verification	Final main 58b0a7a is the reviewed release. Local checks passed on the merged tree; GitHub reported no status checks.	Final-main local checks Passed; CI not reported

Release checklist. T-20 and T-25 passed on final main; all six Lab 4 Issues are Done. PRs #57–#65 are reviewed and merged. The tested main commit is 58b0a7a5556a689ca75c47f9797ebfe84c9cdb3a; exact commands, complete output, and release limits are recorded. GitHub reported no status checks.

Rendered UI contract checklist and evidence limits

Source: docs/lab-04/ui-spec.md. Each item preserves the contract wording and links supporting evidence. Screenshots show state, not every interaction or accessibility criterion.

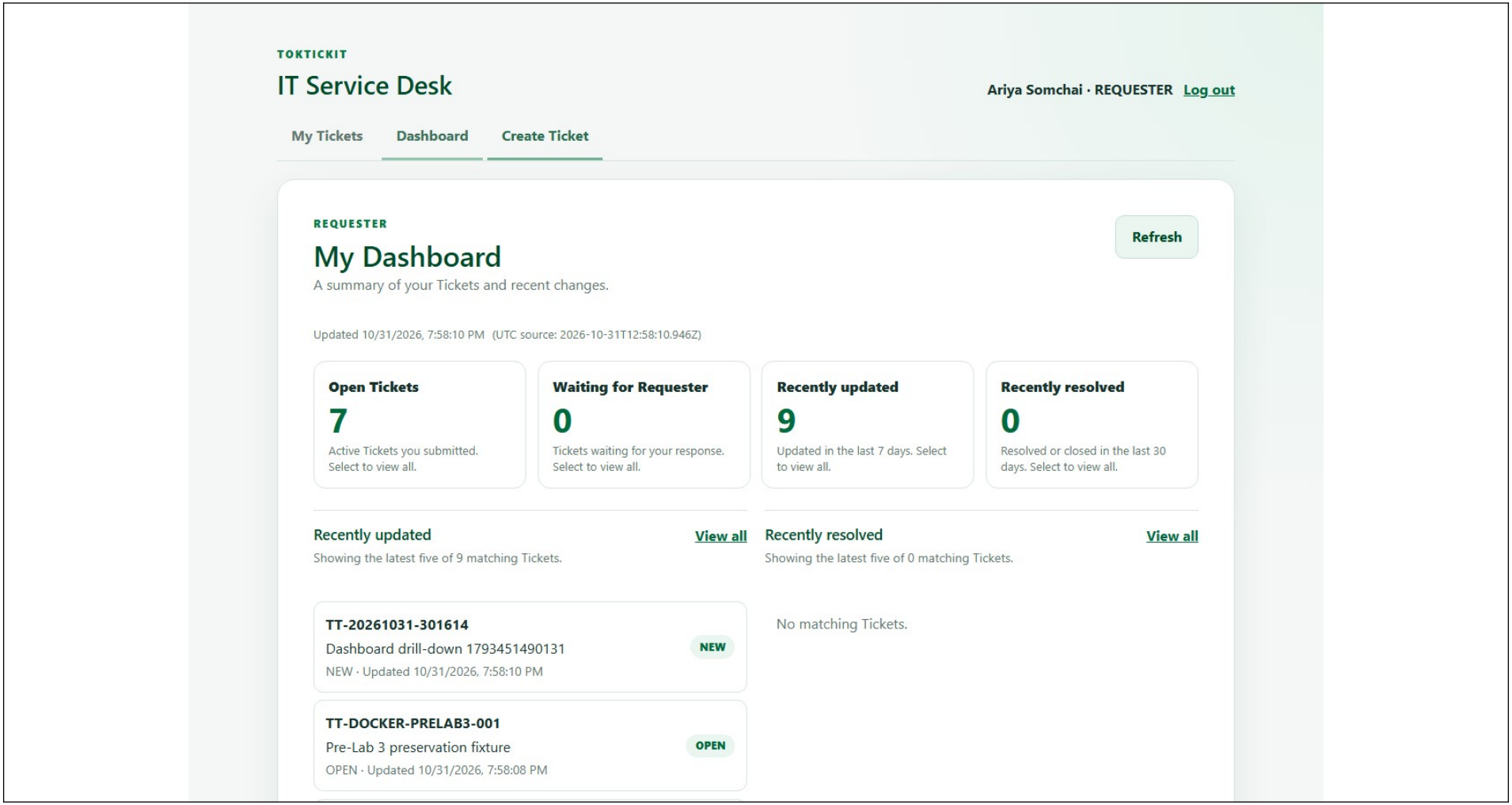
Rendered UI acceptance item	Supporting evidence / limit
1. Requester sees their own action history and no mutation controls.	Requester history / no mutation controls Read-only capture, component test, and server role check.
2. Staff/Admin can complete the action flows and understand performer versus assignee.	Staff/Admin action flow and actor/assignee labels Component tests, multi-action browser journey, and API assignee rules.
3. Illegal transitions are unavailable in the UI and rejected by the backend.	Illegal action and Ticket transitions Action lifecycle matrix, Ticket transition API matrix, and UI legal choices.
4. Resolution blockers are explained and linked to their source actions.	Resolution blockers link to source actions UI blocker test, API gate, and blocked-resolution screenshot.
5. Dashboard counts and drill-down filters agree with server results and show boundary-consistent records.	Dashboard counts / matching drill-downs Server predicate assertions, dashboard UI suites, and browser journey.
6. Loading, empty, validation, authorization, network and concurrency states are understandable and keyboard accessible.	T-20 Passed: dedicated keyboard/focus, named form/region labels, validation alert and aria-invalid focus, reduced motion and 320px overflow checks. Dashboard recovery cases passed for both roles. See release-evidence.md and the appended captures.
7. Narrow layout has no page-level horizontal overflow.	T-20 Passed: dedicated keyboard/focus, named form/region labels, validation alert and aria-invalid focus, reduced motion and 320px overflow checks. Dashboard recovery cases passed for both roles. See release-evidence.md and the appended captures.

Responsive evidence: 36 source captures overall, including 12 new 320px views. The report embeds its original 24 plus four dedicated T-20 captures. Focus, labels/ARIA validation, reduced motion and overflow assertions passed. T-25 records the Done board, actual reviews, prompts, commands and results. No standalone accessibility audit or CI pass is claimed.

Requester dashboard metrics and recent Tickets

Source captures from the final PR6 Playwright run. Open the full screenshot index

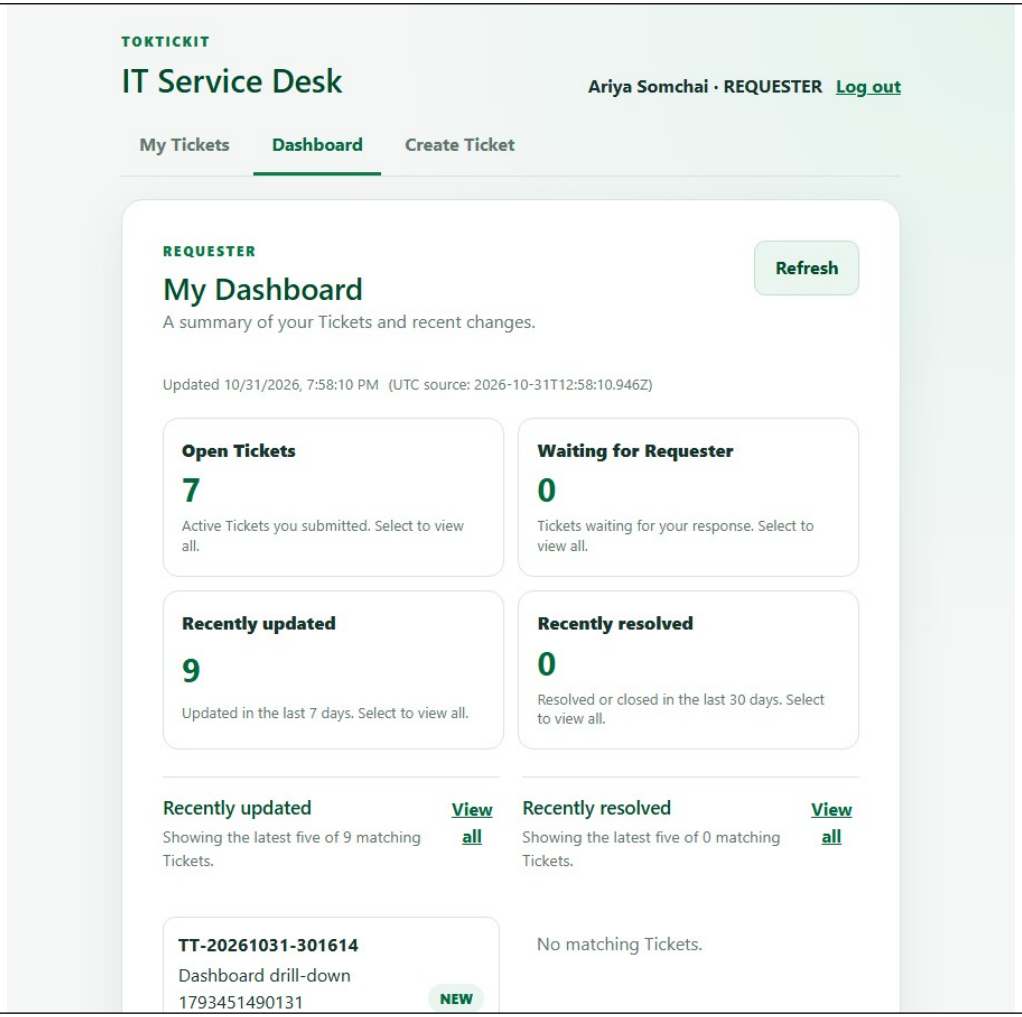
Requester dashboard metrics and recent Tickets — Desktop 1280×900



View full image

Automated test: e2e/lab-04/dashboards.spec.ts:6

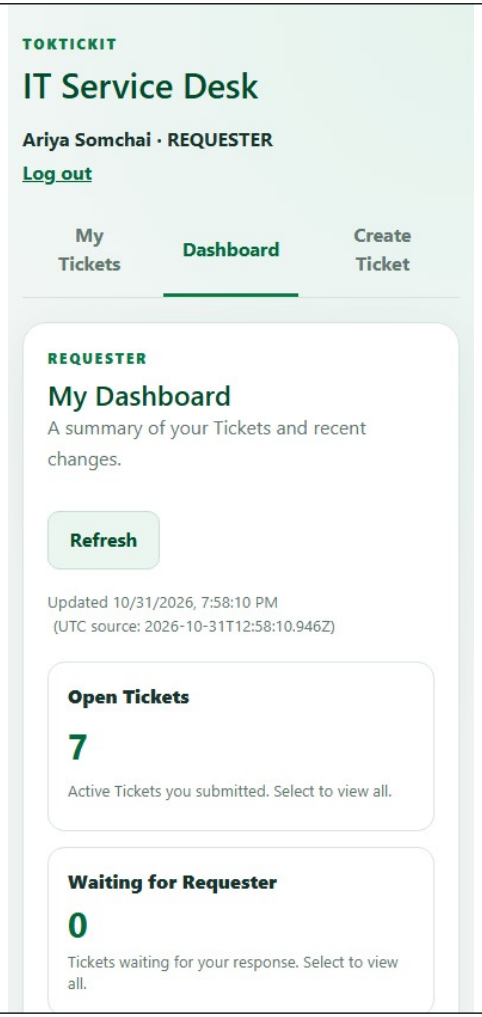
Requester dashboard metrics and recent Tickets — Tablet 900×900



View full image

Automated test: e2e/lab-04/dashboards.spec.ts:6

Requester dashboard metrics and recent Tickets — Mobile 390×844



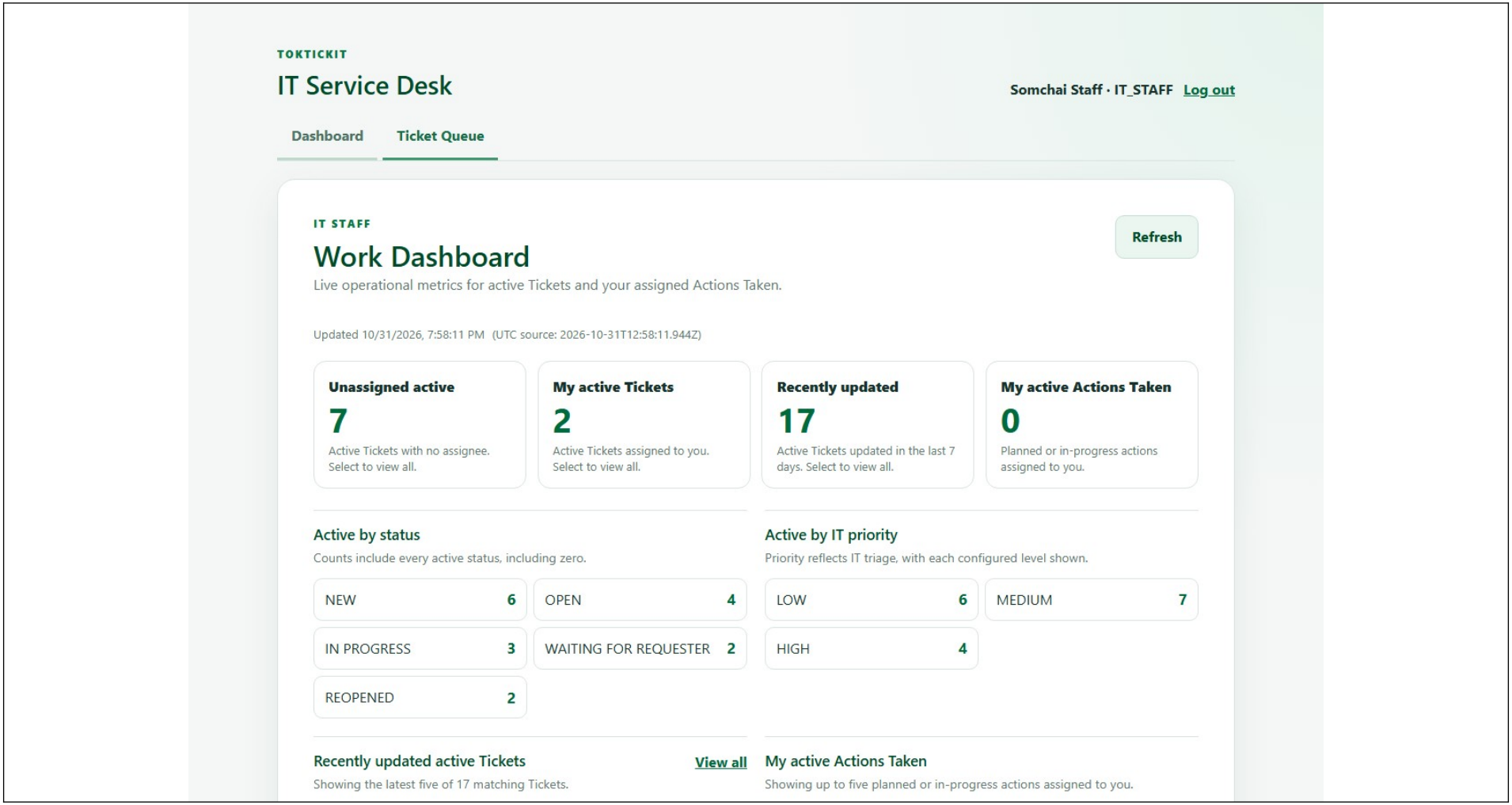
View full image

Automated test: e2e/lab-04/dashboards.spec.ts:6

Staff dashboard metrics and active Ticket counts

Source captures from the final PR6 Playwright run. Open the full screenshot index

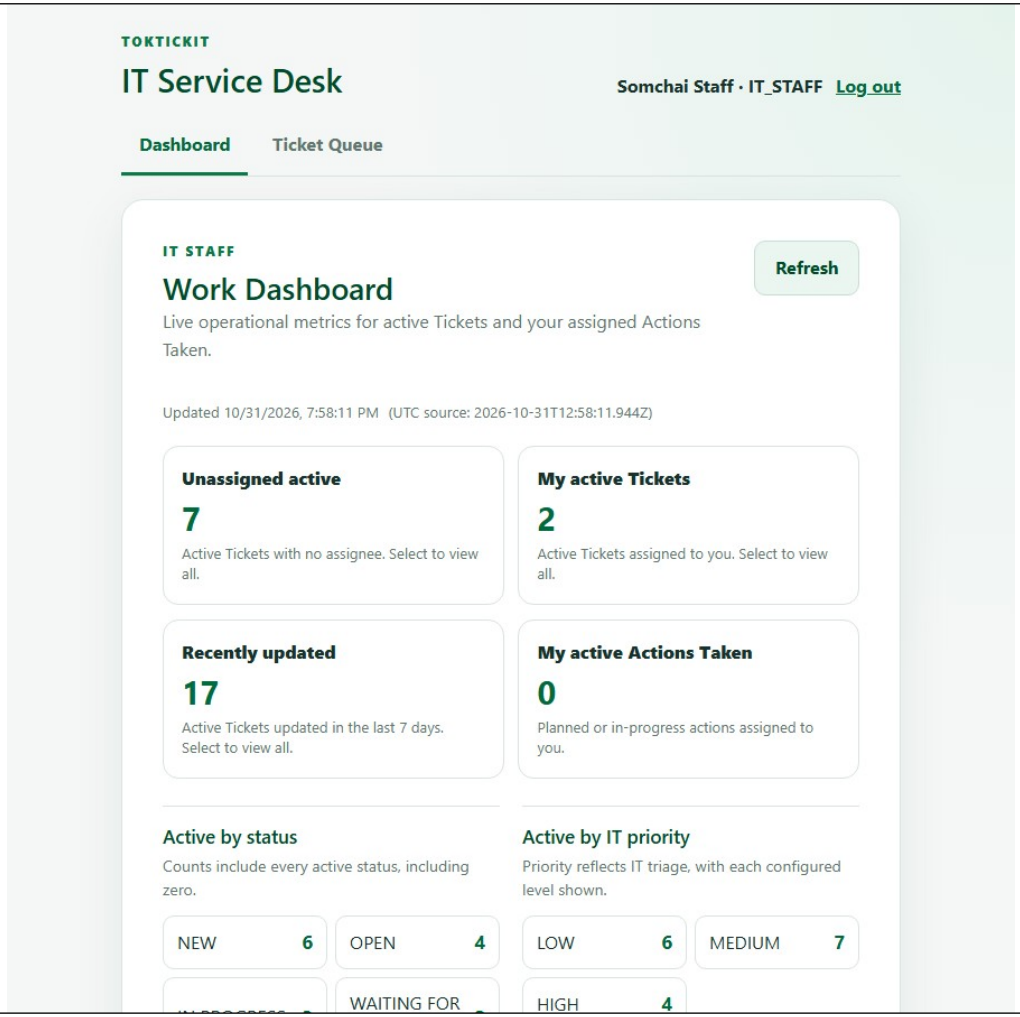
Staff dashboard metrics and active Ticket counts — Desktop 1280×900



View full image

Automated test: e2e/lab-04/dashboards.spec.ts:6

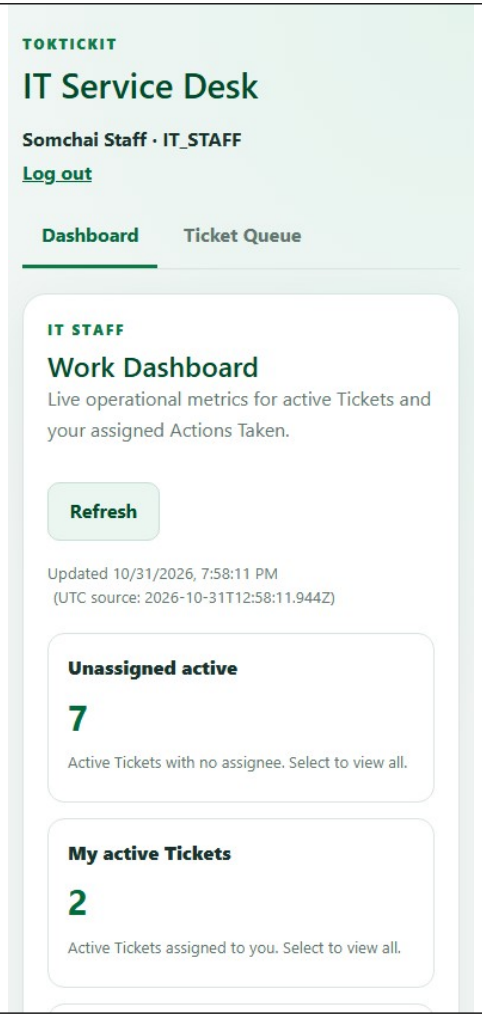
Staff dashboard metrics and active Ticket counts — Tablet 900×900



View full image

Automated test: e2e/lab-04/dashboards.spec.ts:6

Staff dashboard metrics and active Ticket counts — Mobile 390×844



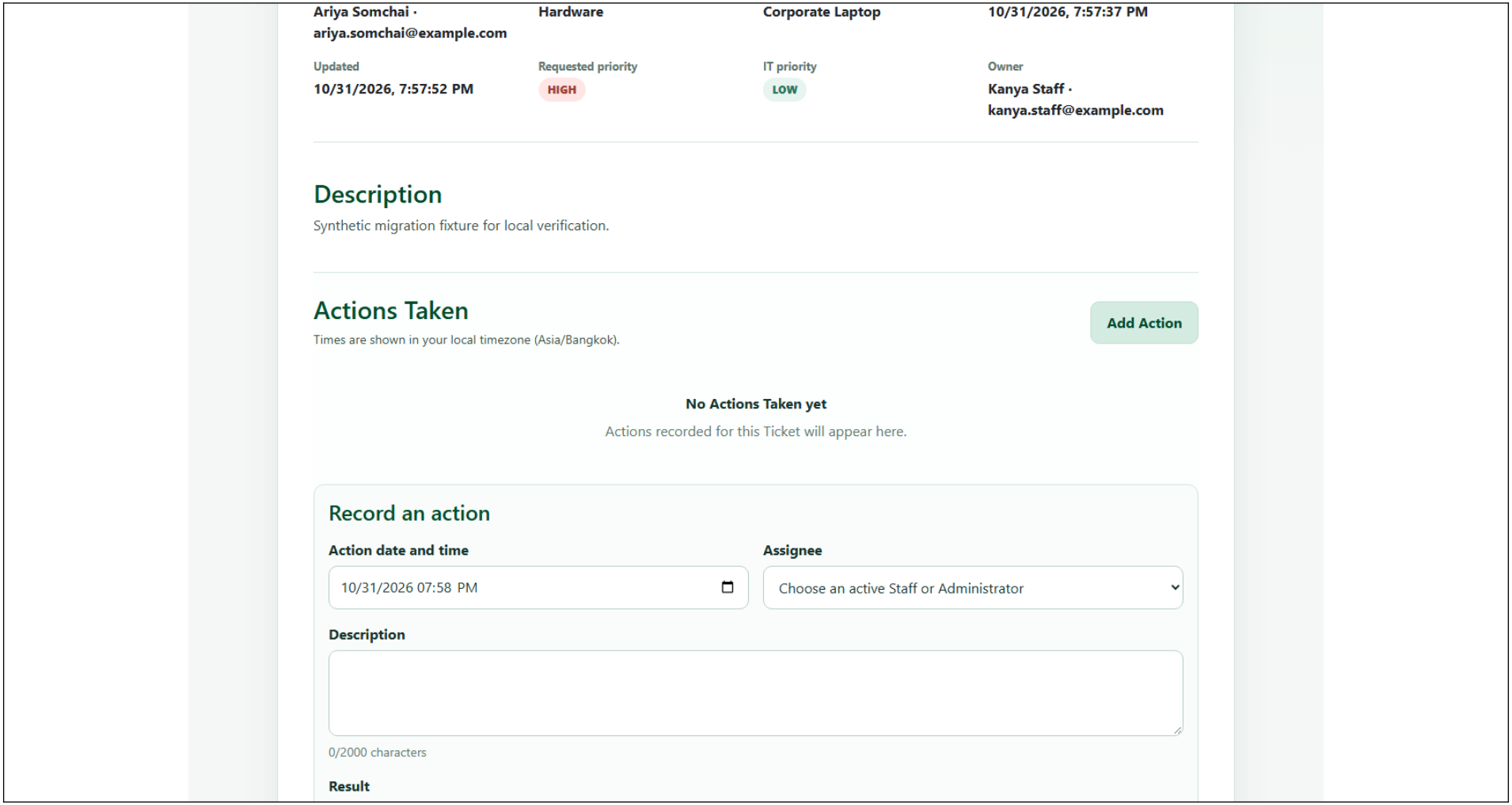
View full image

Automated test: e2e/lab-04/dashboards.spec.ts:6

Staff action entry form

Source captures from the final PR6 Playwright run. Open the full screenshot index

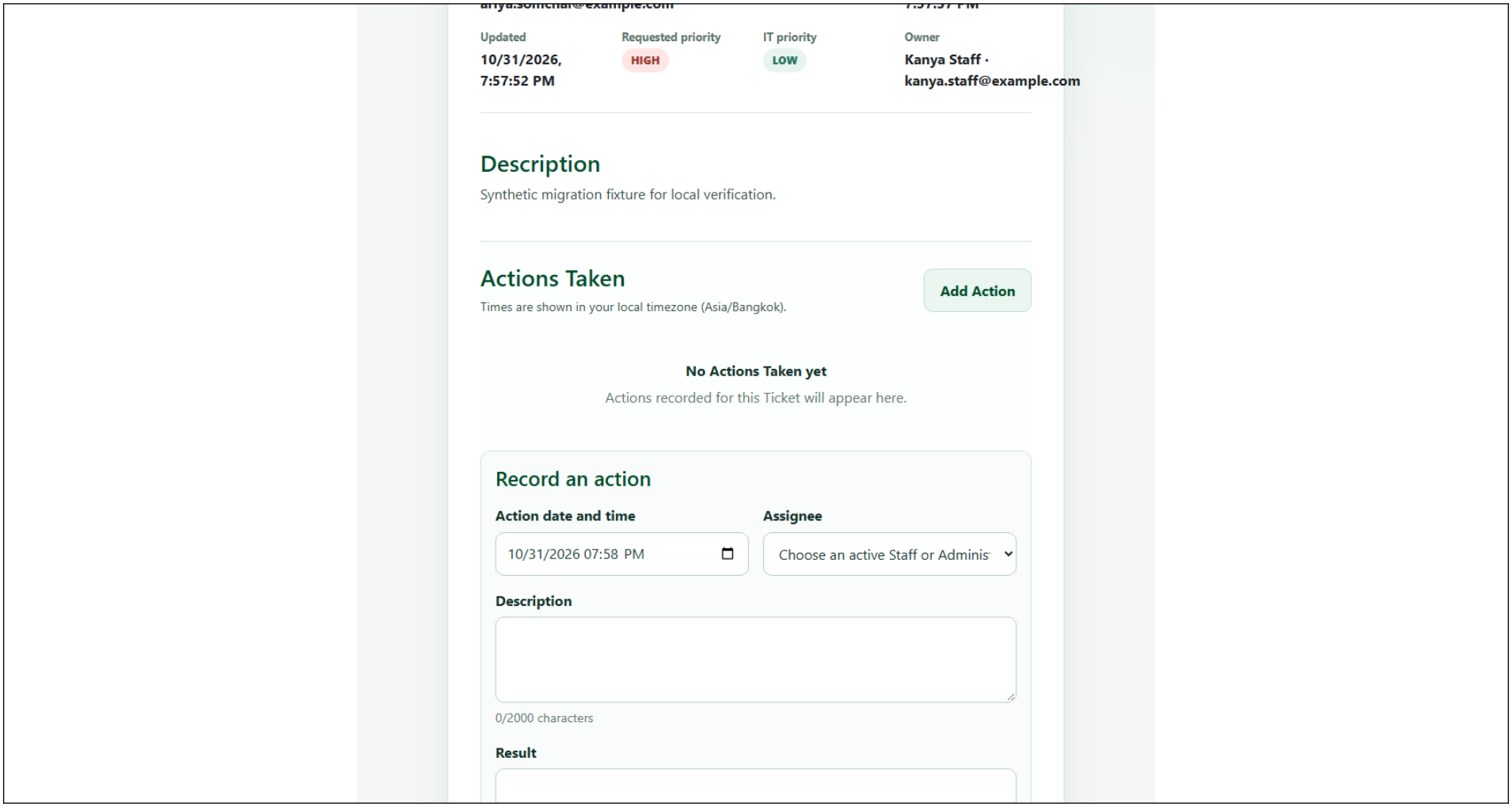
Staff action entry form — Desktop 1280×900



View full image

Automated test: client/tests/lab-04/ActionsTaken.test.tsx:105

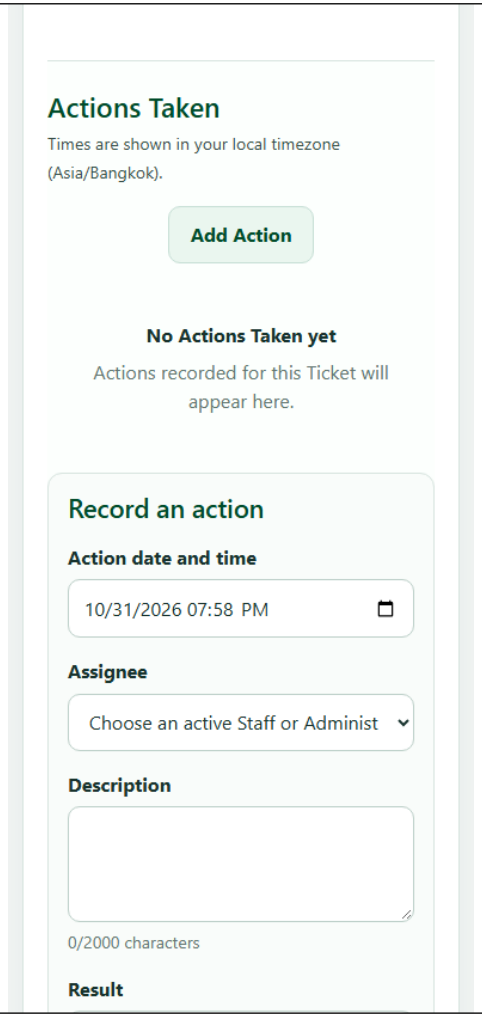
Staff action entry form — Tablet 900×900



View full image

Automated test: client/tests/lab-04/ActionsTaken.test.tsx:105

Staff action entry form — Mobile 390×844



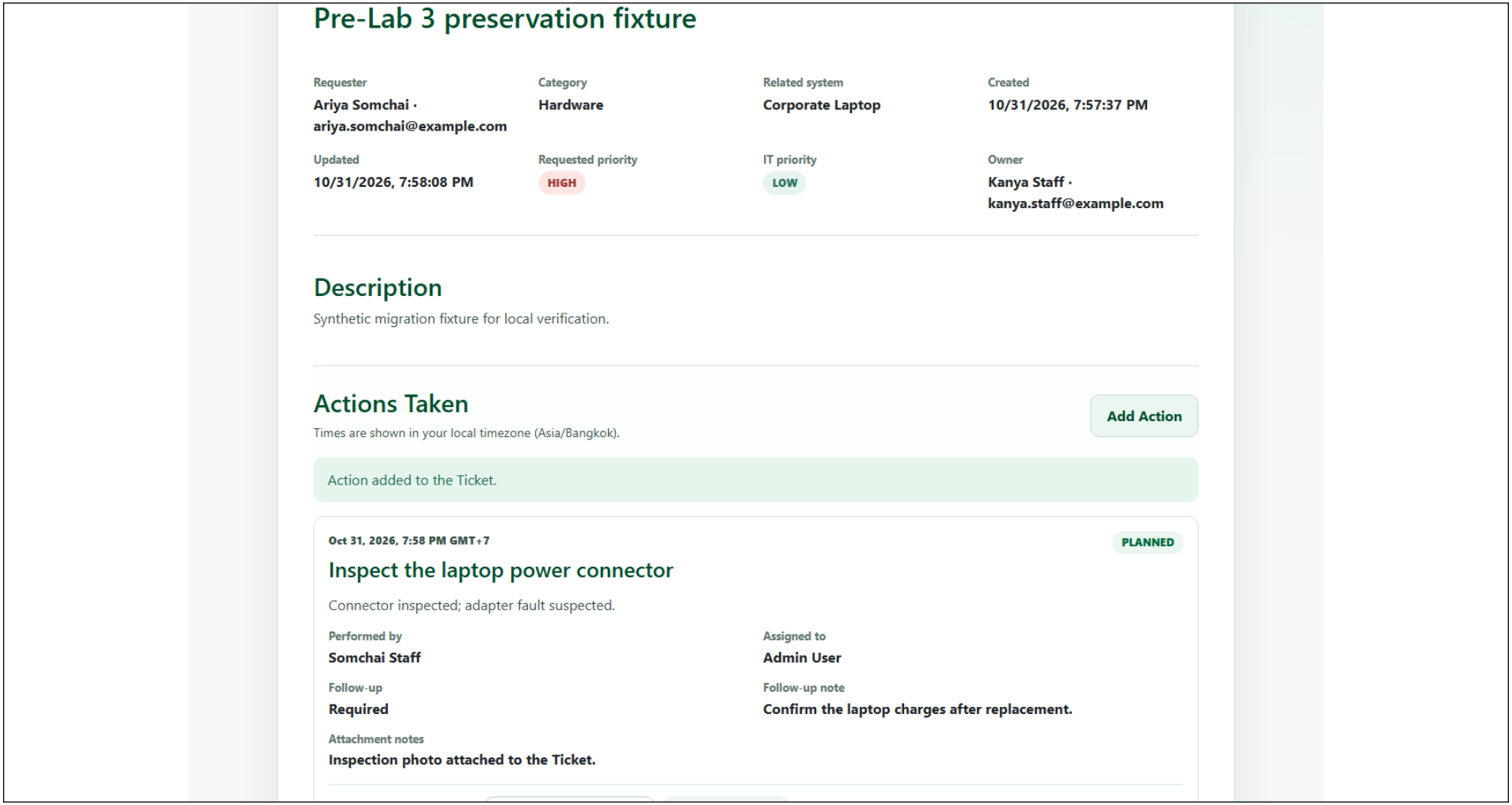
View full image

Automated test: client/tests/lab-04/ActionsTaken.test.tsx:105

Staff history with multiple recorded actions

Source captures from the final PR6 Playwright run. Open the full screenshot index

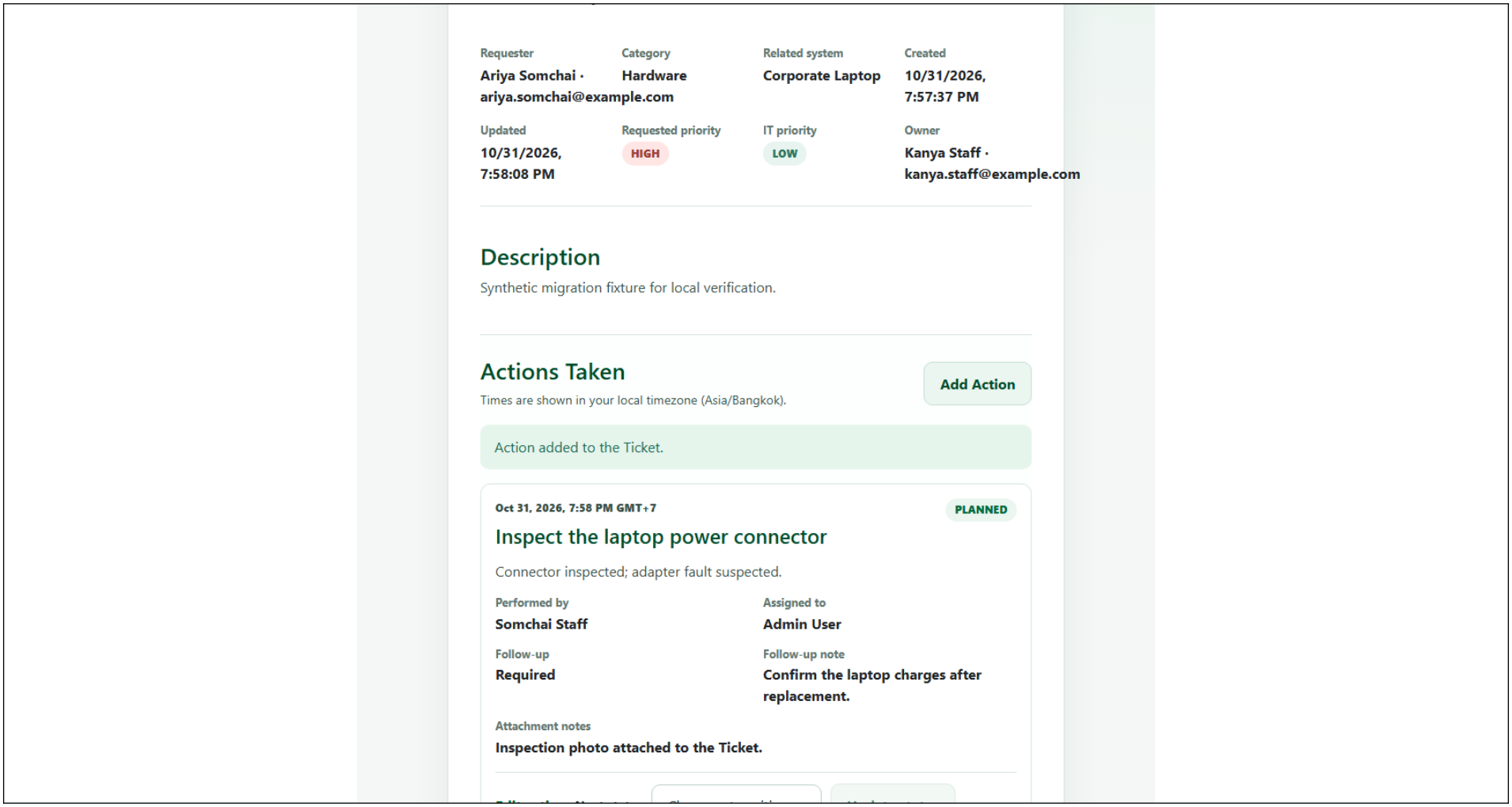
Staff history with multiple recorded actions — Desktop 1280×900



View full image

Automated test: e2e/lab-04/actions-taken-flow.spec.ts:27

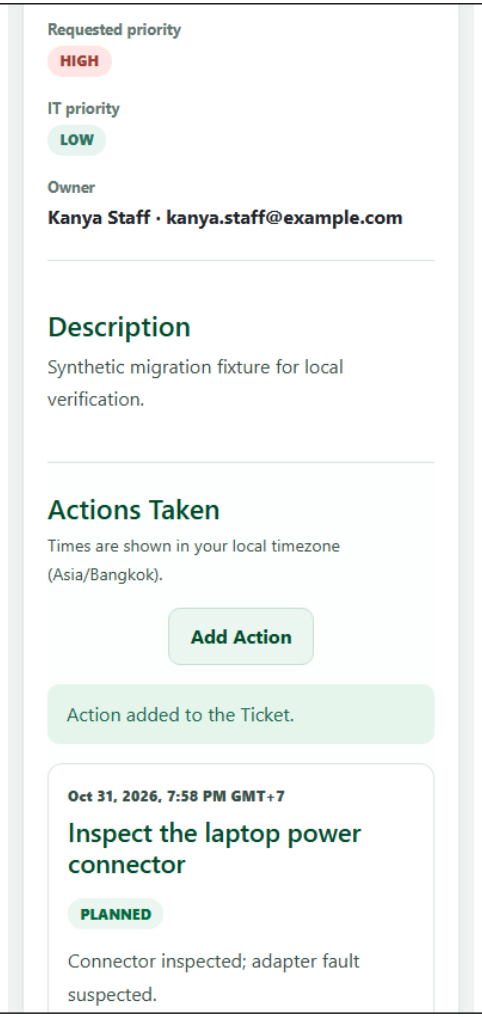
Staff history with multiple recorded actions — Tablet 900×900



View full image

Automated test: e2e/lab-04/actions-taken-flow.spec.ts:27

Staff history with multiple recorded actions — Mobile 390×844



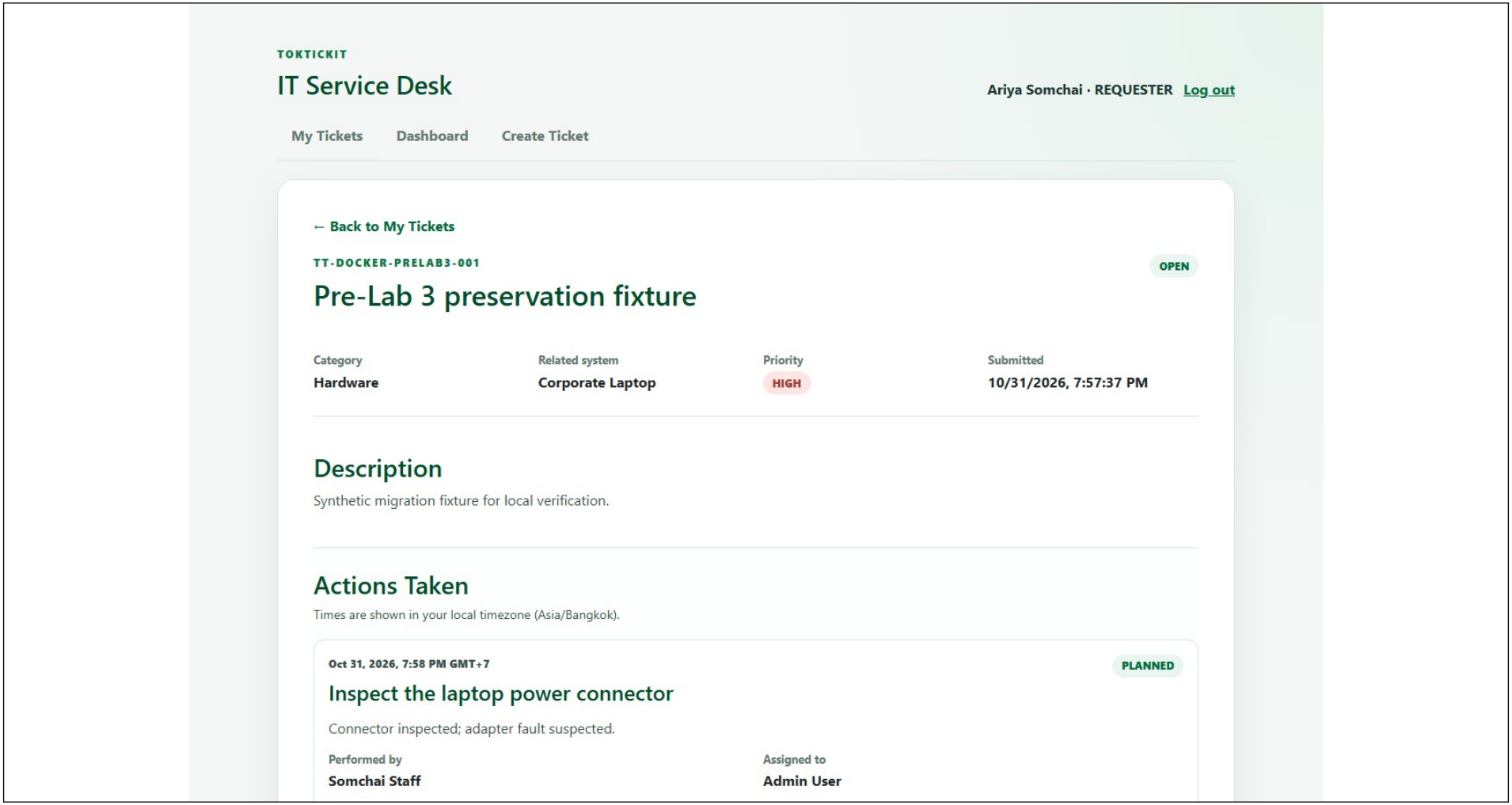
View full image

Automated test: e2e/lab-04/actions-taken-flow.spec.ts:27

Requester read-only Actions Taken history

Source captures from the final PR6 Playwright run. Open the full screenshot index

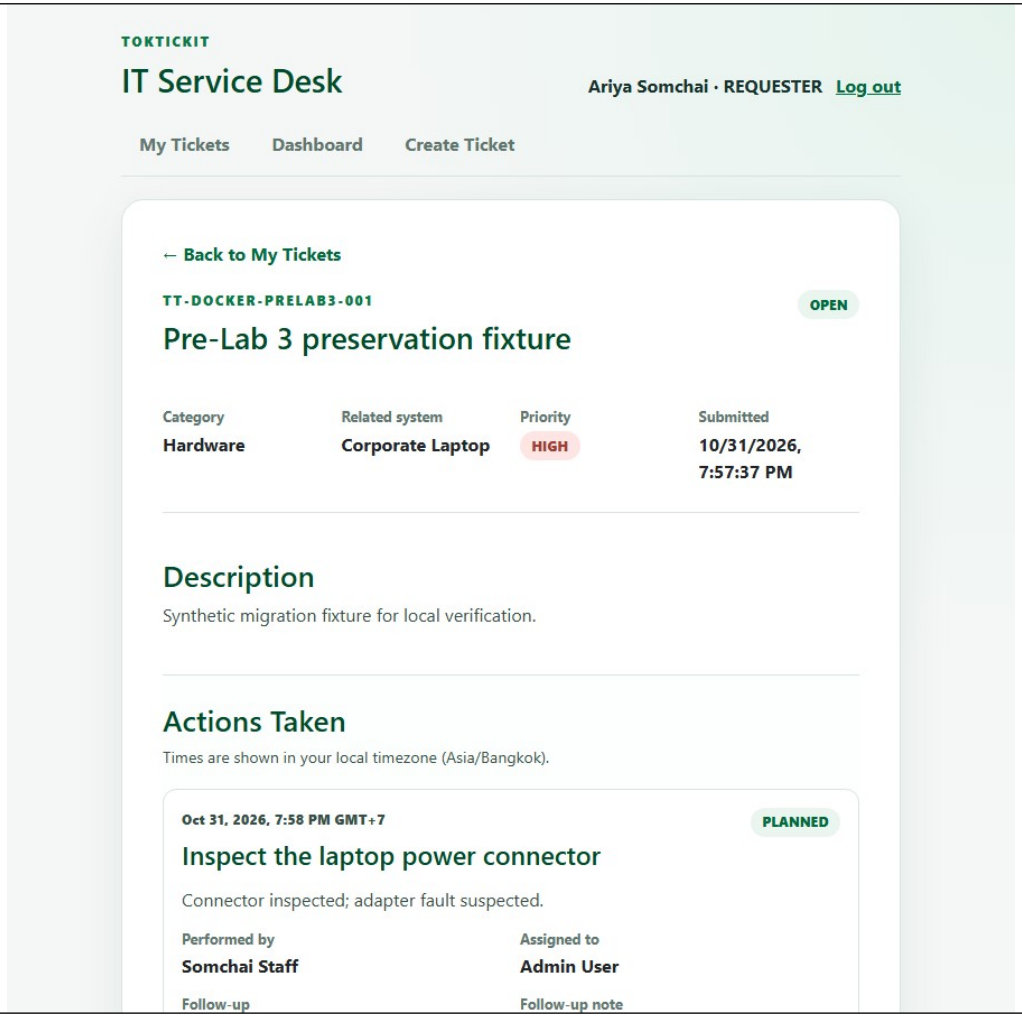
Requester read-only Actions Taken history — Desktop 1280×900



View full image

Automated test: client/tests/lab-04/ActionsTaken.test.tsx:43

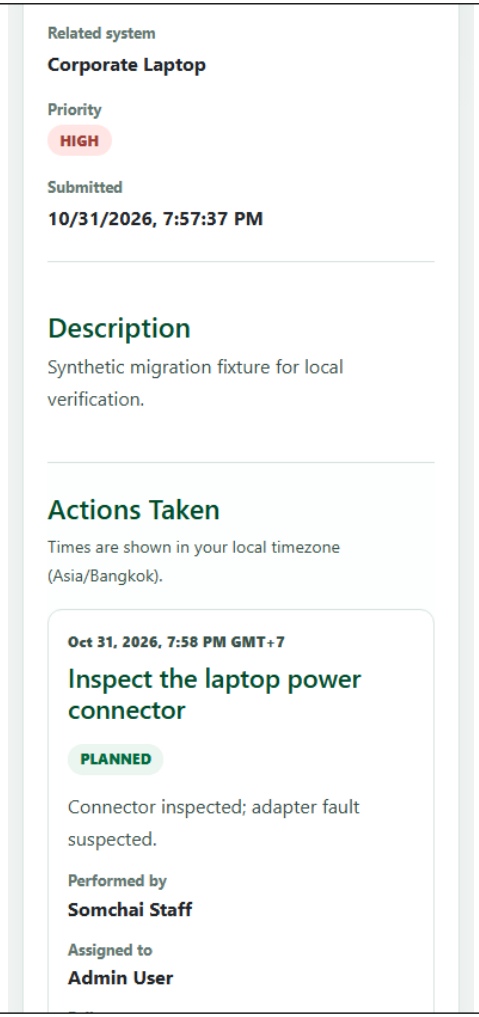
Requester read-only Actions Taken history — Tablet 900×900



View full image

Automated test: client/tests/lab-04/ActionsTaken.test.tsx:43

Requester read-only Actions Taken history — Mobile 390×844



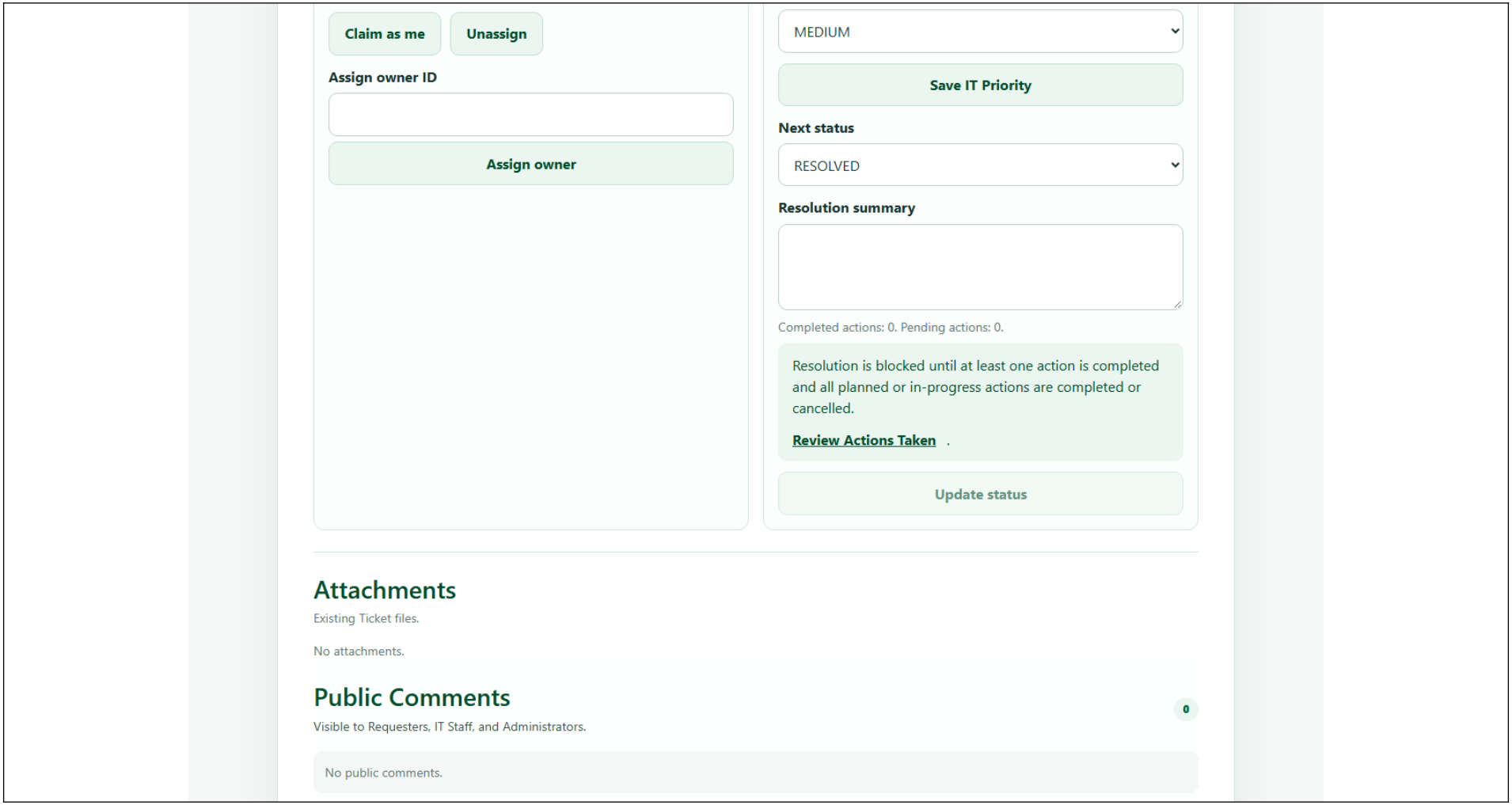
View full image

Automated test: client/tests/lab-04/ActionsTaken.test.tsx:43

Resolution blocked before completed work

Source captures from the final PR6 Playwright run. Open the full screenshot index

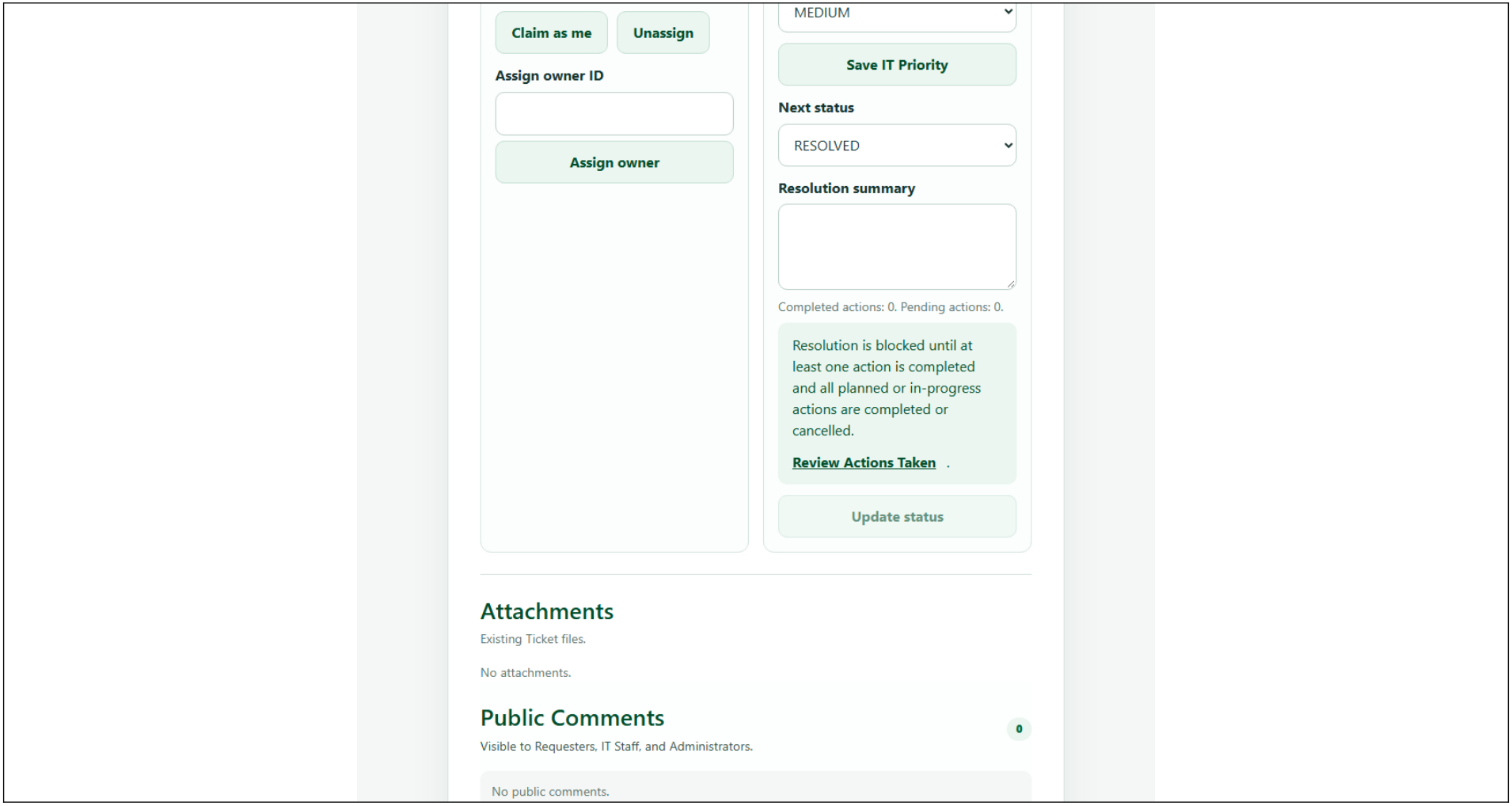
Resolution blocked before a completed action — Desktop 1280×900



View full image

Automated test: client/tests/lab-04/TicketWorkflow.test.tsx:55

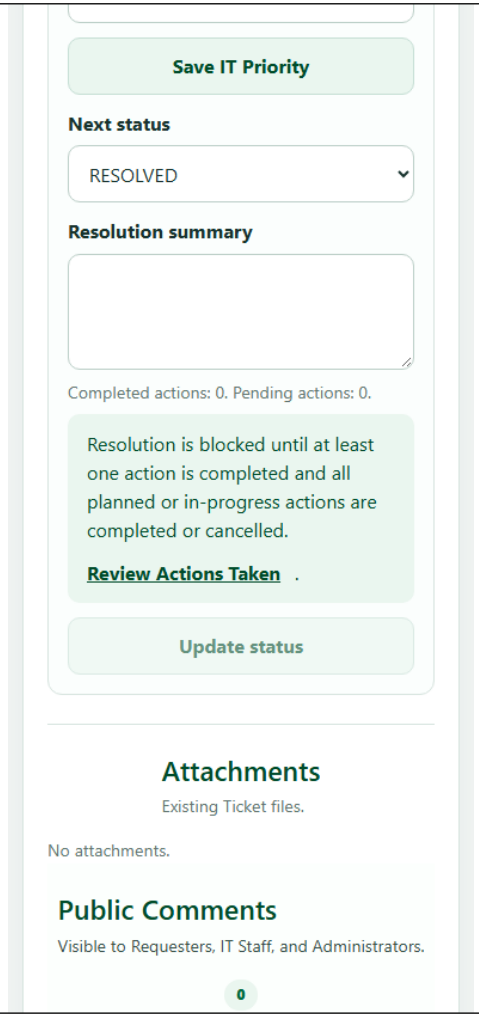
Resolution blocked before a completed action — Tablet 900×900



View full image

Automated test: client/tests/lab-04/TicketWorkflow.test.tsx:55

Resolution blocked before a completed action — Mobile 390×844



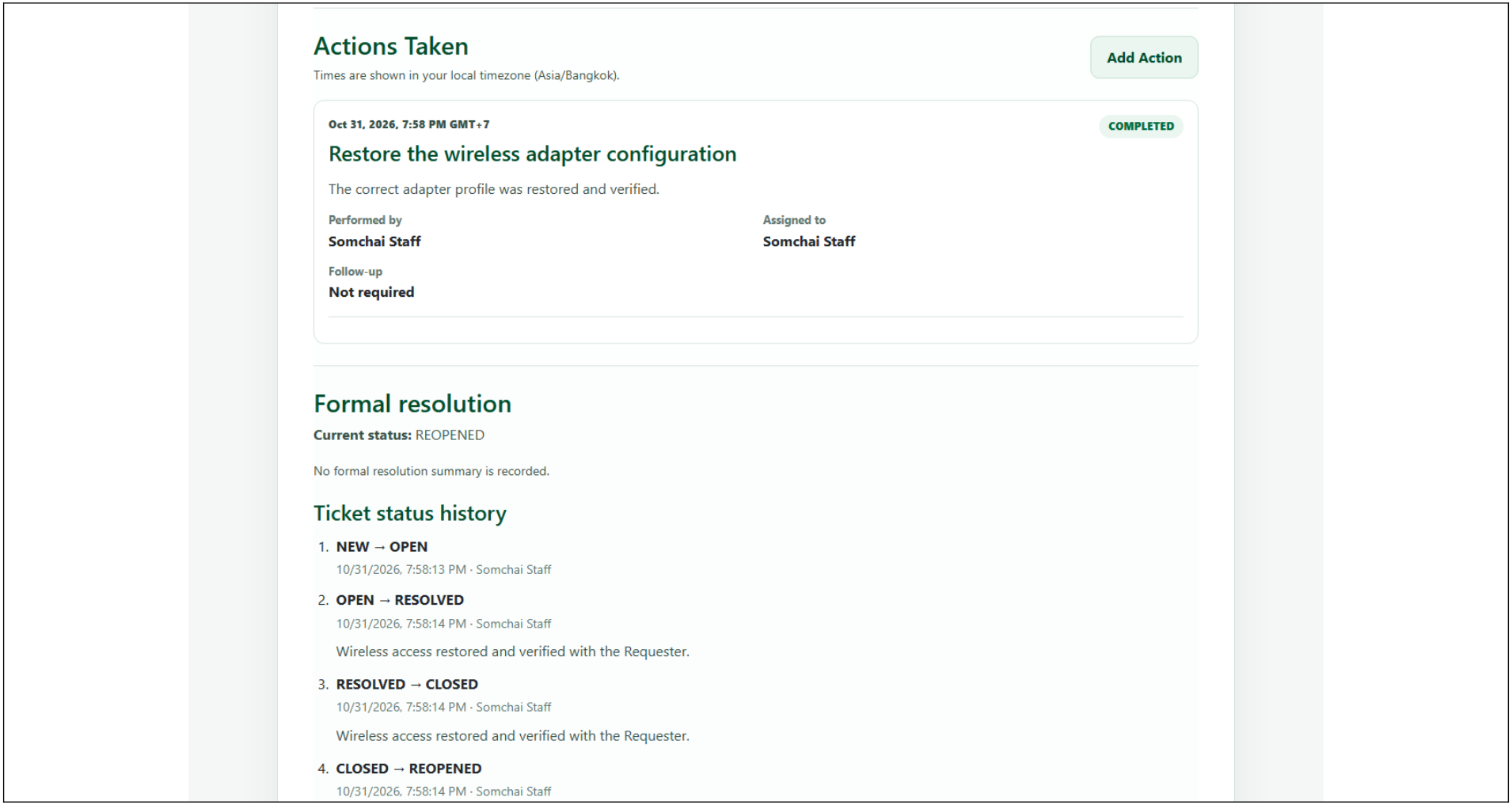
View full image

Automated test: client/tests/lab-04/TicketWorkflow.test.tsx:55

Staff view after reopening a Ticket

Source captures from the final PR6 Playwright run. Open the full screenshot index

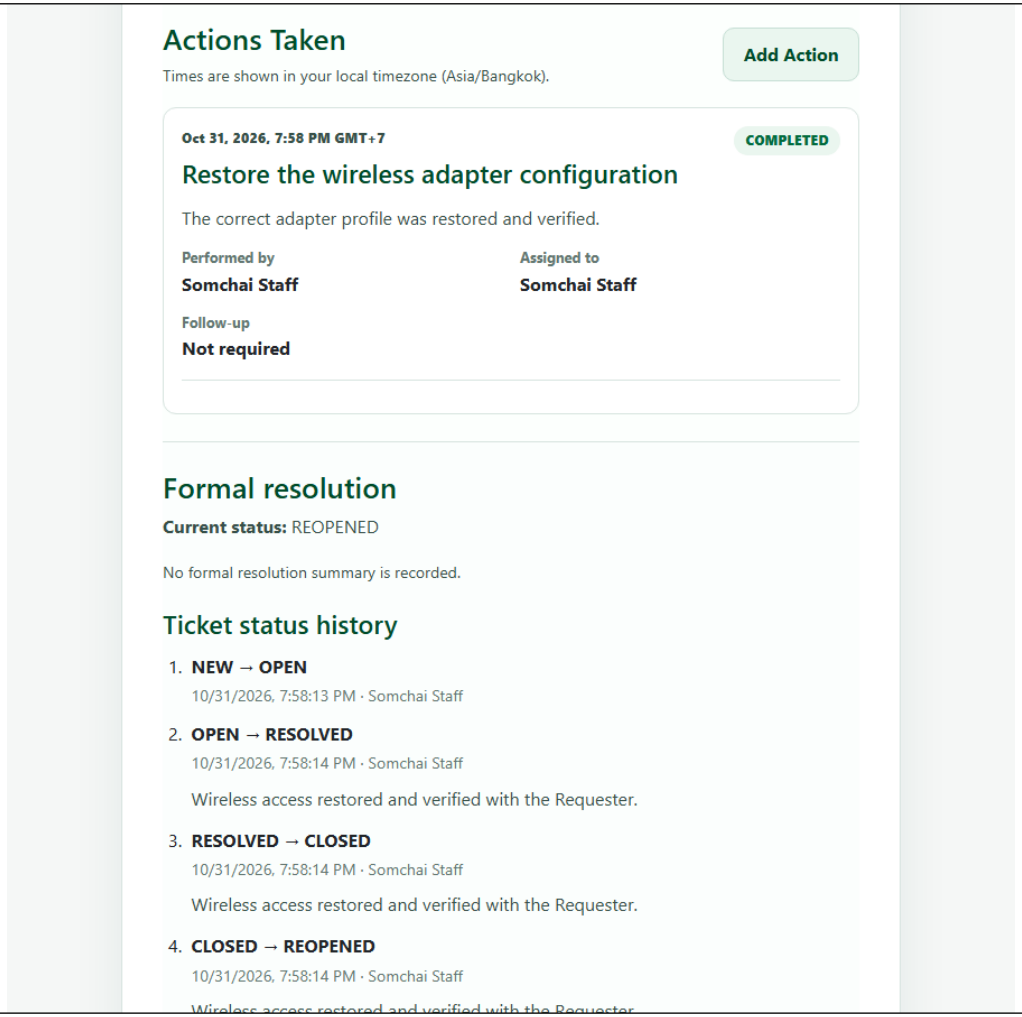
Staff view after reopening a Ticket — Desktop 1280×900



View full image

Automated test: e2e/lab-04/ticket-resolution.spec.ts:6

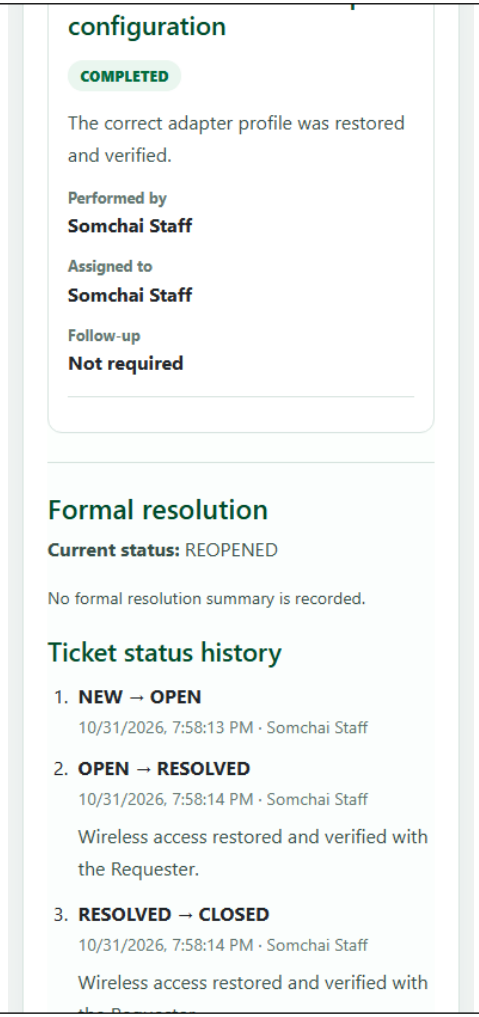
Staff view after reopening a Ticket — Tablet 900×900



View full image

Automated test: e2e/lab-04/ticket-resolution.spec.ts:6

Staff view after reopening a Ticket — Mobile 390×844



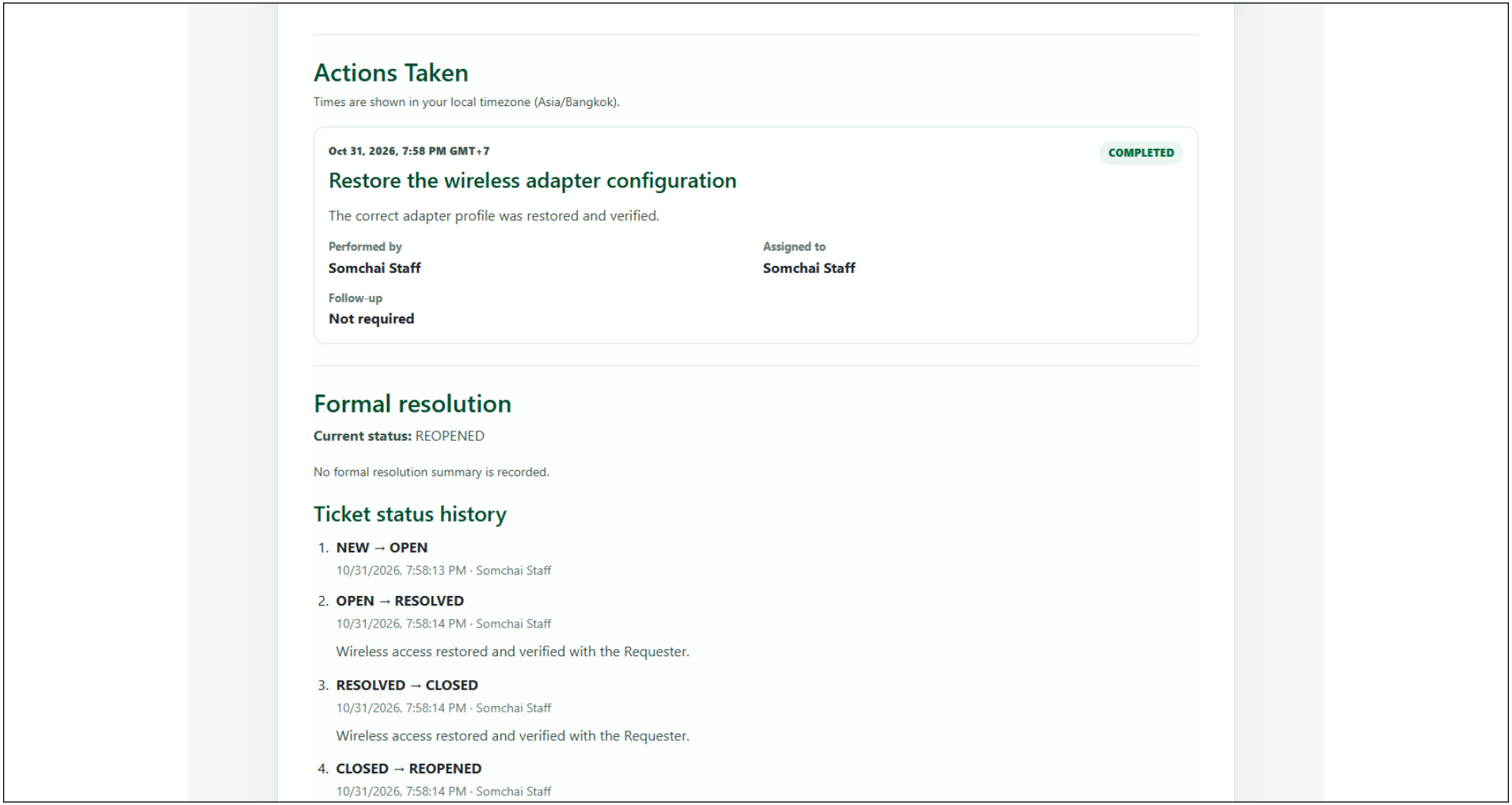
View full image

Automated test: e2e/lab-04/ticket-resolution.spec.ts:6

Requester resolution and Ticket history

Source captures from the final PR6 Playwright run. Open the full screenshot index

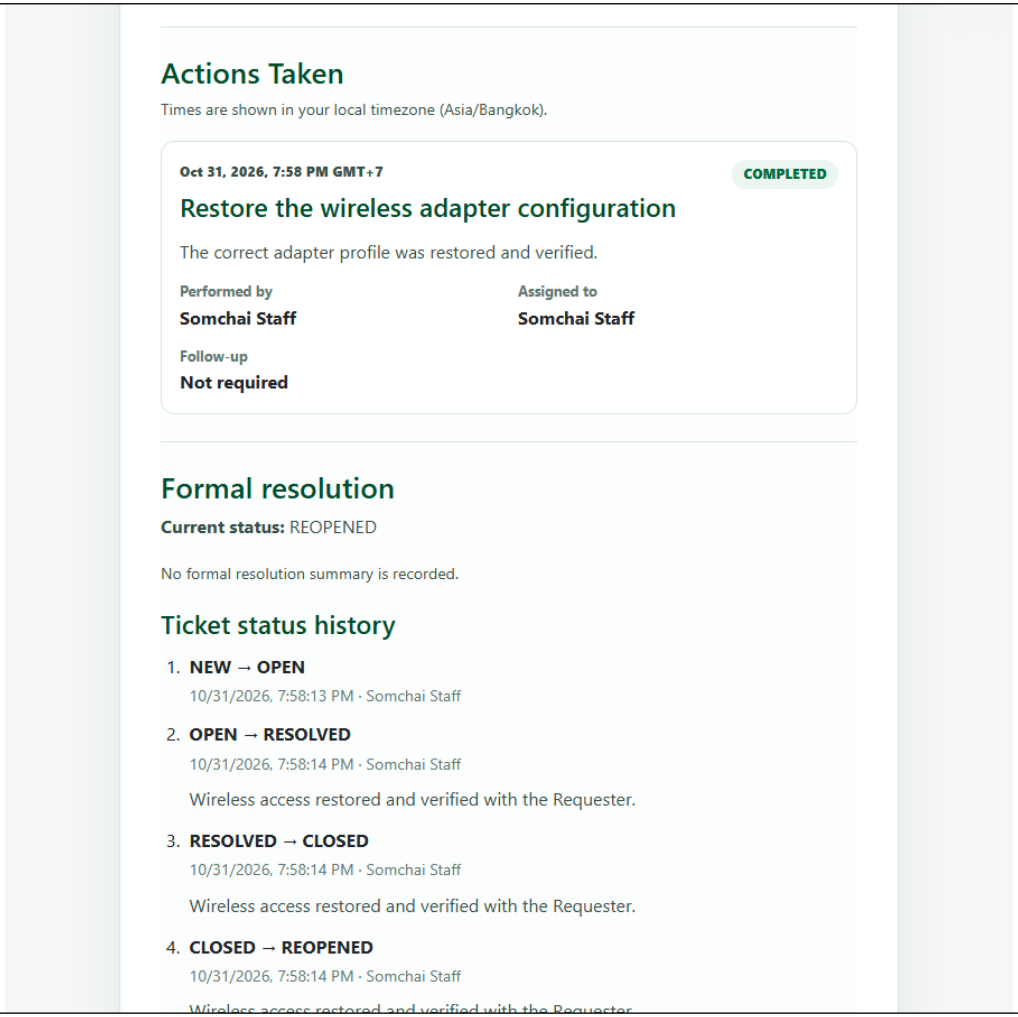
Requester resolution and Ticket history — Desktop 1280×900



View full image

Automated test: e2e/lab-04/ticket-resolution.spec.ts:6

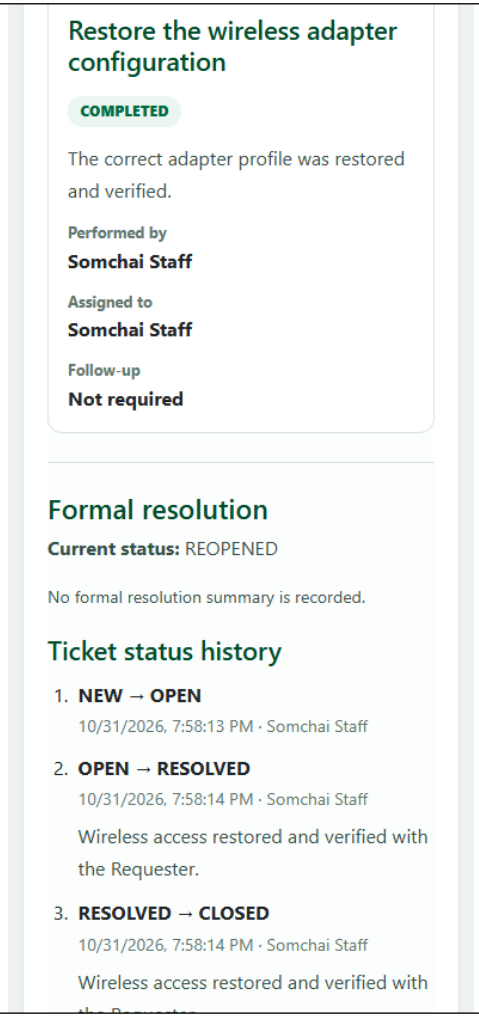
Requester resolution and Ticket history — Tablet 900×900



View full image

Automated test: e2e/lab-04/ticket-resolution.spec.ts:6

Requester resolution and Ticket history — Mobile 390×844



View full image

Automated test: e2e/lab-04/ticket-resolution.spec.ts:6

Completed T20 and T25 release evidence

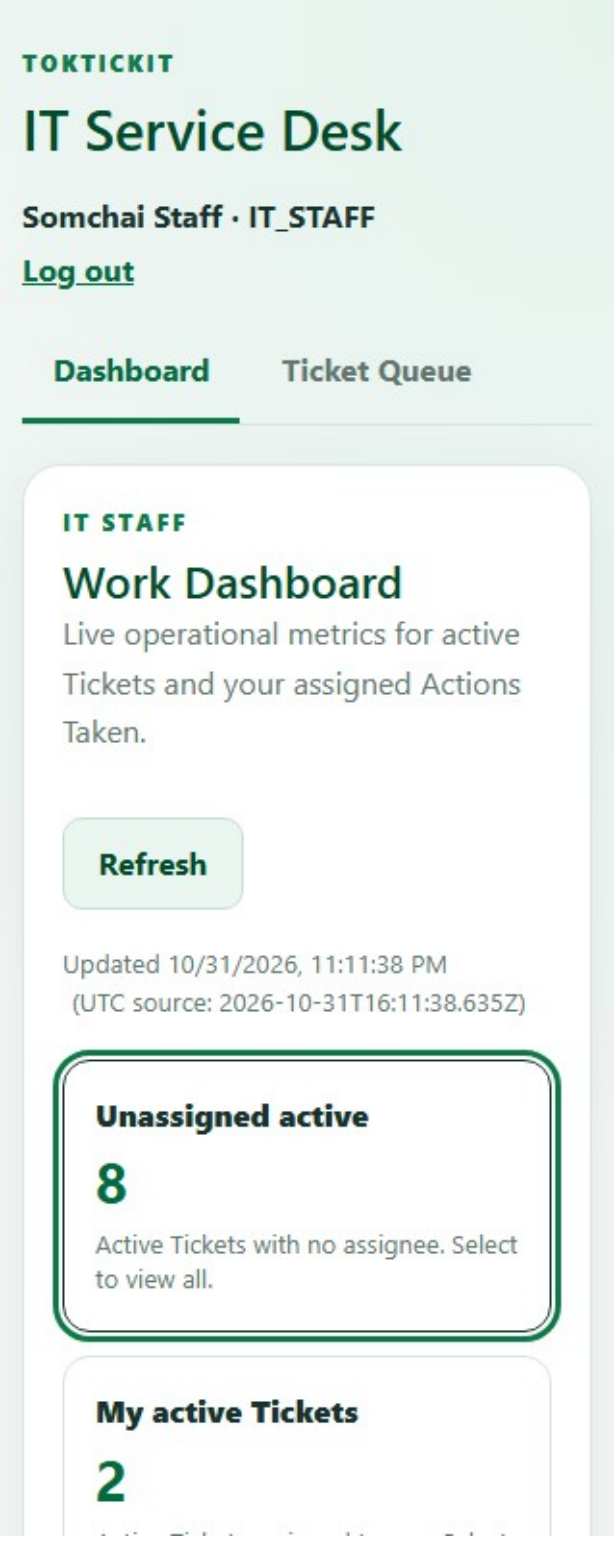
T-20 and T-25 passed on final main commit 58b0a7a5556a689ca75c47f9797ebfe84c9cdb3a. The merged tree passed 306 server tests, 66 client tests, 22 browser tests, both builds, Prisma validation and production health. Issues #51–#56 are verified in Done. GitHub reported no status checks.

The board was inspected read-only at https://github.com/users/book6349/projects/4/views/1. Backlog, Specified, Started, PR Review and Fixing each had zero items; Done contained Issues #51–#56. The release checklist and source links are in docs/lab-04/release-evidence.md.

Checklist item	Evidence	Result
Implementation identity	58b0a7a5556a689ca75c47f9797ebfe84c9cdb3a	Passed
Regression	306 server / 66 client / 22 browser; builds and Prisma valid	Passed
T20	Keyboard/focus, labels/ARIA validation, reduced motion, 320px overflow	Passed
Dashboard failure handling	Both roles: loading, network/forbidden failure, retry, retained metrics and filter recovery	Passed
Review record	Punge089 comment, author reply, approval, and merge for PRs #57–#65	Passed
Issue and board state	All six Lab 4 Issues closed and in Done	Passed
AI use	Six actual prompt excerpts and reflection preserved	Passed
Release boundary	PR #65 merged; main verified at 58b0a7a. GitHub reported no status checks.	Passed

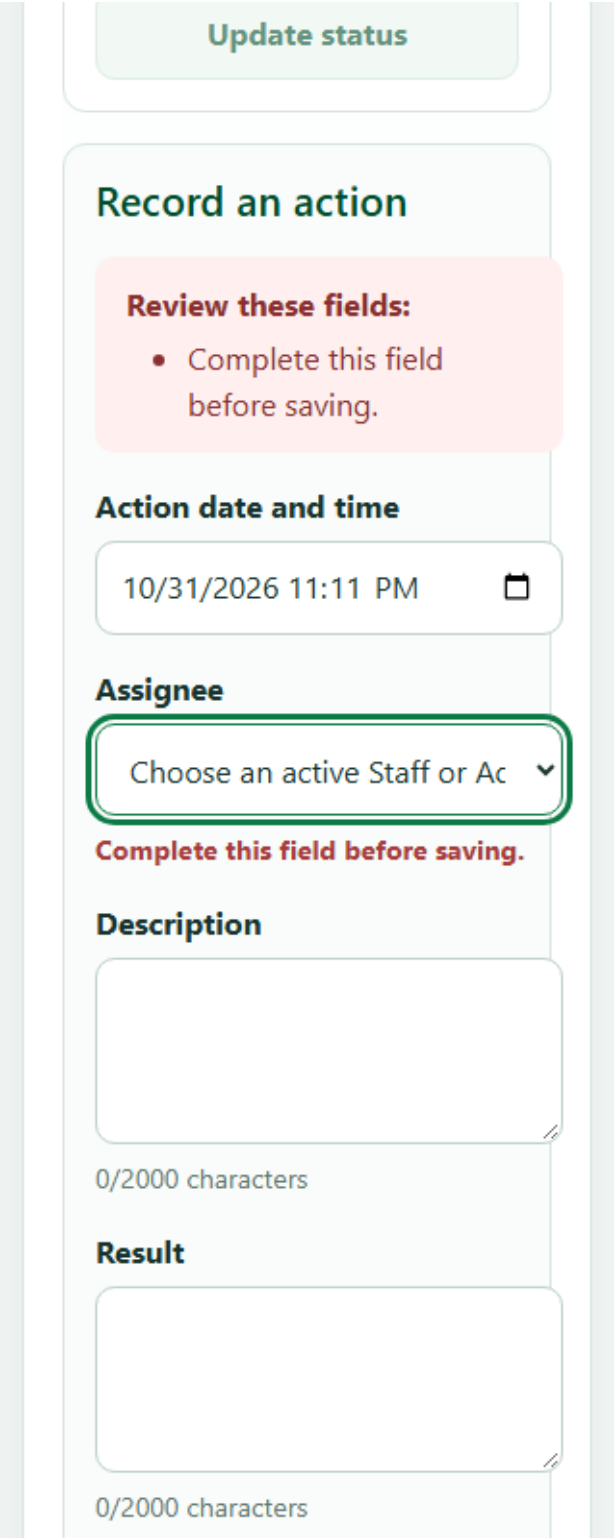
Staff dashboard keyboard focus at 320 pixels

Capture committed with PR #65 head 20b4e0a; that tree matches merged main 58b0a7a. The final-main browser run reran the no-horizontal-overflow assertion at 320×800.



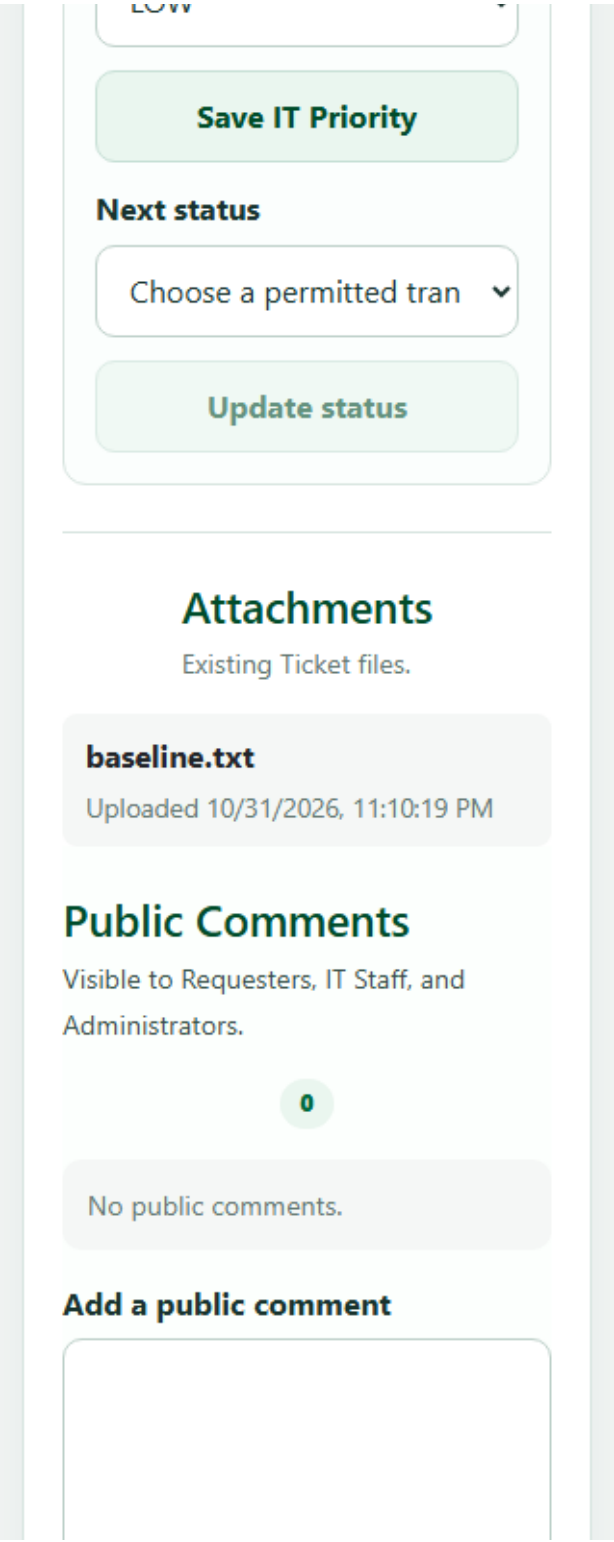
Actions Taken validation focus at 320 pixels

Capture committed with PR #65 head 20b4e0a; that tree matches merged main 58b0a7a. The final-main browser run reran the no-horizontal-overflow assertion at 320×800.



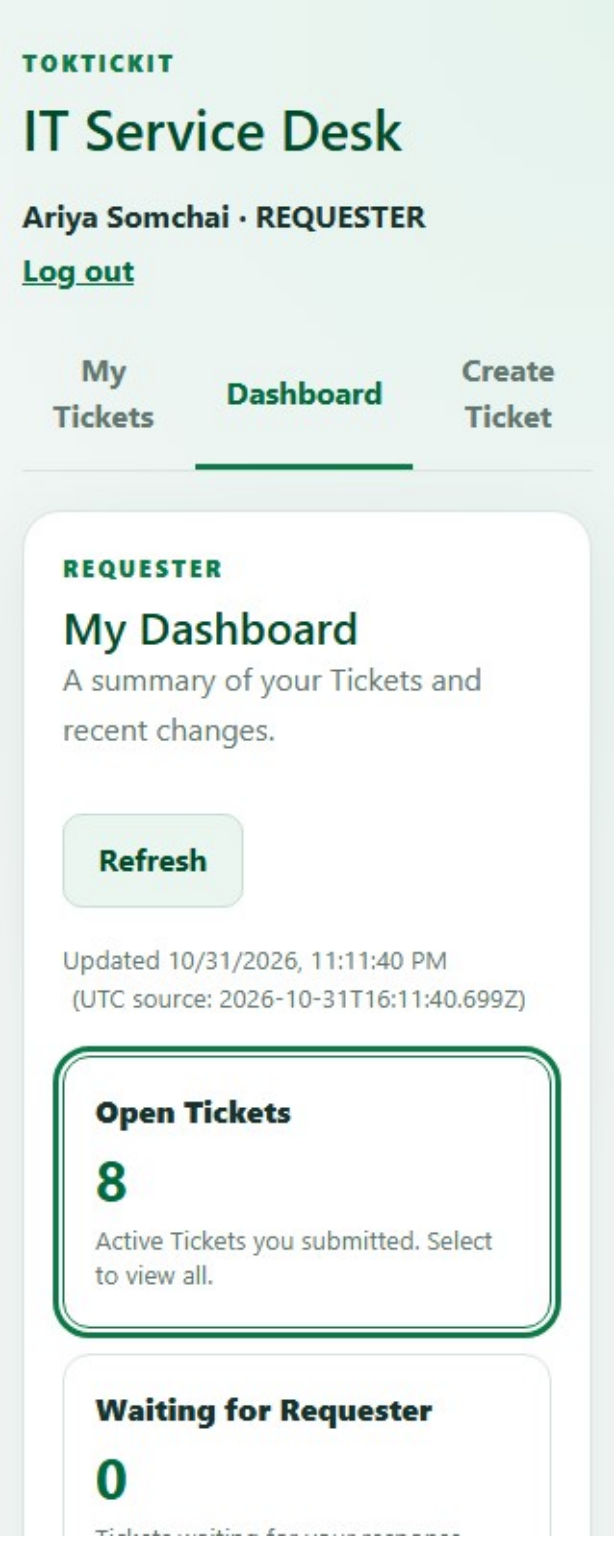
Staff Ticket workflow at 320 pixels

Capture committed with PR #65 head 20b4e0a; that tree matches merged main 58b0a7a. The final-main browser run reran the no-horizontal-overflow assertion at 320×800.



Requester dashboard keyboard focus at 320 pixels

Capture committed with PR #65 head 20b4e0a; that tree matches merged main 58b0a7a. The final-main browser run reran the no-horizontal-overflow assertion at 320×800.



Final-main verification after PR #65

PR #65 merged to main at 58b0a7a5556a689ca75c47f9797ebfe84c9cdb3a on 1 November 2026 (Asia/Bangkok, UTC+07:00). The merge tree 65731368fb2ad8c27e86da34248906a1a5881ee6 is identical to the tested PR head tree at 20b4e0a7698d3eb5527cd9eb85d36e2ce8c9daeb, so the local verification below covers the exact merged source tree.

Server: 306 tests in 23 files, including 37 PostgreSQL-backed checks; client: 66 tests in 18 files; browser: 22 Playwright tests on one worker (16 Lab 3, 2 Lab 2, 4 Lab 4) against a fresh isolated database. Server and client production builds, Prisma schema validation, migration/seed assertions, and compiled-server health passed.

Limits: GitHub returned no status checks for this revision. No GitHub CI pass, full WCAG audit, manual screen-reader evaluation, standalone axe scan, or performance benchmark is claimed.

Review source: PR #65 · merge commit · review comment · author reply · approval

Server, client, production builds, and Prisma validation

```
Final-main verification: validation, 2026-11-01 Asia/Bangkok
Revision: 58b0a7a5556a689ca75c47f9797ebfe84c9cdb3a (merge PR #65)
Tree: 65731368fb2ad8c27e86da34248906a1a5881ee6
Source tree compared equal to tested PR #65 tree at 20b4e0a7698d3eb5527cd9eb85d36e2ce8c9daeb.

Command: server: LAB4_DB_TEST=1 npm test -- --reporter=dot

> toktickit-server@1.0.0 test
> vitest run --reporter=dot



 RUN  v2.1.9 C:/Users/Jitvoottikrai/OneDrive/เดสก์ท็อป/SoftwareEn/LAB1/Lab1_Starter_Scaffold/toktickit/LAB4/toktickit-main-
verify-6e66c2c/server

 ✓ tests/lab-03/migration-regression.test.ts (2 tests) 12ms
 ✓ tests/lab-02/validation.unit.test.ts (3 tests) 5ms
 ✓ tests/lab-03/auth.unit.test.ts (4 tests) 393ms
 ✓ tests/lab-03/e2e-fixtures.unit.test.ts (11 tests) 625ms
   ✓ Lab 3 disposable E2E preparation safeguards > uses actual email-based IDs, preserves accounts, and populates stable
realistic Queue fixtures 375ms
 ✓ tests/lab-02/ticket-detail.api.test.ts (2 tests) 63ms
 ✓ tests/lab-02/requester-context.api.test.ts (2 tests) 70ms
 ✓ tests/lab-02/my-tickets.api.test.ts (2 tests) 75ms
 ✓ tests/lab-04/staff-dashboard.api.test.ts (3 tests) 114ms
 ✓ tests/lab-04/requester-dashboard.api.test.ts (4 tests) 120ms
 ✓ tests/lab-03/authorization.api.test.ts (3 tests) 145ms
 ✓ tests/lab-02/create-ticket.api.test.ts (2 tests) 182ms
 ✓ tests/lab-02/attachments.api.test.ts (3 tests) 162ms
 ✓ tests/lab-03/comments-notes.api.test.ts (8 tests) 330ms
 ✓ tests/lab-03/staff-queue.api.test.ts (38 tests) 347ms
 ✓ tests/lab-02/ticket-number.unit.test.ts (1 test) 5ms
 ✓ tests/lab-02/reference-data.api.test.ts (2 tests) 49ms
 ✓ tests/lab-04/ticket-workflow.api.test.ts (72 tests) 837ms
 ✓ tests/lab-01/categories.test.ts (1 test) 35ms
 ✓ tests/lab-01/health.test.ts (1 test) 35ms
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts (89 tests) 992ms
 ✓ tests/lab-03/auth.api.test.ts (7 tests) 1127ms
   ✓ Lab 3 authentication contract > requires CSRF and returns safe user data on login 384ms
   ✓ Lab 3 authentication contract > gates normal routes until password change and revokes the old session 387ms
 ✓ tests/lab-03/users-admin.api.test.ts (9 tests) 2792ms
   ✓ Lab 3 Administrator User Management API > lists safe users with case-insensitive search and one role filter 381ms
   ✓ Lab 3 Administrator User Management API > rejects non-Administrators and invalid role filters 512ms
   ✓ Lab 3 Administrator User Management API > creates an active user with a hashed initial password and forced change 308ms
   ✓ Lab 3 Administrator User Management API > rejects duplicate email and invalid create input safely 462ms
   ✓ Lab 3 Administrator User Management API > edits permitted account fields and resets the initial password 328ms
stdout | tests/lab-04/actions-taken.api.test.ts > Lab 4 Actions Taken with disposable PostgreSQL
Disposable database: toktickit_lab4_prep_1793514957050

 ✓ tests/lab-04/actions-taken.api.test.ts (37 tests) 10264ms
   ✓ Lab 4 Actions Taken with disposable PostgreSQL > rolls back action, event and Ticket version when persistence fails
523ms

 Test Files  23 passed (23)
      Tests  306 passed (306)
   Start at  13:35:55
   Duration  12.32s (transform 1.87s, setup 0ms, collect 22.57s, tests 18.78s, environment 6ms, prepare 7.43s)

Server test exit code: 0

Command: client: npm test -- --reporter=dot

> toktickit-client@1.0.0 test
> vitest run --reporter=dot



 RUN  v2.1.9 C:/Users/Jitvoottikrai/OneDrive/เดสก์ท็อป/SoftwareEn/LAB1/Lab1_Starter_Scaffold/toktickit/LAB4/toktickit-main-
verify-6e66c2c/client

 ✓ tests/lab-01/App.test.tsx (3 tests) 345ms
   ✓ App > renders the TokTickIT sign-in heading after checking the session 331ms
 ✓ tests/lab-03/ui-style.test.tsx (2 tests) 343ms
   ✓ Lab 3 Zen Green styling and responsive rules > renders accessible sign-in controls with the primary action hierarchy
337ms
 ✓ tests/lab-02/RequesterContext.test.tsx (1 test) 483ms
```

```
   ✓ Lab 2 requester context > restores the authenticated Requester and ignores stale development-selector state 481ms
 ✓ tests/lab-03/ChangePassword.test.tsx (2 tests) 940ms
   ✓ Lab 3 mandatory Change Password > blocks the normal shell until a valid replacement password is saved 841ms
 ✓ tests/lab-03/Login.test.tsx (3 tests) 1106ms
   ✓ Lab 3 Login > shows the login form and enters the authenticated Requester shell without a selector 580ms
 ✓ tests/lab-02/ui-style.test.tsx (2 tests) 1153ms
   ✓ Lab 2 Zen Green UI conventions > exposes the required theme tokens and visible button hierarchy 858ms
 ✓ tests/lab-02/AttachmentSection.test.tsx (1 test) 1510ms
   ✓ Lab 2 AttachmentSection > allows upload and removal for active files while hiding actions for removed files 1501ms
 ✓ tests/lab-03/StaffTicketQueue.test.tsx (2 tests) 1547ms
   ✓ Lab 3 IT Staff Ticket Queue > loads the authenticated Staff Queue and sends the documented filters 1209ms
   ✓ Lab 3 IT Staff Ticket Queue > distinguishes an empty Queue from a filtered no-result state 329ms
 ✓ tests/lab-04/RequesterDashboard.test.tsx (6 tests) 1654ms
   ✓ Lab 4 Requester dashboard UI > shows UTC context, full counts, accessible definitions and matching owned-Ticket drill-
downs 460ms
   ✓ Lab 4 Requester dashboard UI > retains visible zero metrics and lets the user return from an empty drill-down 411ms
 ✓ tests/lab-02/MyTickets.test.tsx (2 tests) 1702ms
   ✓ Lab 2 My Tickets > sends the selected filters and sort controls to the requester-scoped API 1183ms
   ✓ Lab 2 My Tickets > supports pagination and distinguishes a filtered no-results state 515ms
 ✓ tests/lab-03/RequesterWorkflow.test.tsx (3 tests) 2000ms
   ✓ Lab 3 Requester ticket regression > adds a public comment to the owned ticket and never renders Internal Notes 1206ms
   ✓ Lab 3 Requester ticket regression > shows the Requester resolution indication after the API confirms it 472ms
   ✓ Lab 3 Requester ticket regression > renders comment markup as text without creating executable HTML 319ms
 ✓ tests/lab-04/StaffDashboard.test.tsx (6 tests) 2199ms
   ✓ Lab 4 Staff dashboard UI > shows all status and IT-priority values and drills down through the dashboard filter API
714ms
   ✓ Lab 4 Staff dashboard UI > keeps performer and assignee distinct and opens the linked Ticket detail 326ms
   ✓ Lab 4 Staff dashboard UI > announces initial failure Network request failed. and recovers through Retry 357ms
   ✓ Lab 4 Staff dashboard UI > announces initial loading and keeps the last metrics after a failed refresh 341ms
 ✓ tests/lab-02/RequesterTicketDetail.test.tsx (2 tests) 2123ms
   ✓ Lab 2 owned Ticket Detail > renders an owned detail as read-only and navigates back to My Tickets 1812ms
   ✓ Lab 2 owned Ticket Detail > shows a safe error when the detail is not owned or no longer exists 307ms
 ✓ tests/lab-03/UserManagement.test.tsx (3 tests) 2231ms
   ✓ Lab 3 Administrator User Management > lists users, applies search and role filters, and keeps the direct detail entry
point 974ms
   ✓ Lab 3 Administrator User Management > creates a user with clear initial-password behavior 507ms
   ✓ Lab 3 Administrator User Management > edits an account and exposes reset initial password as a separate action 747ms
 ✓ tests/lab-02/CreateTicket.test.tsx (4 tests) 2328ms
   ✓ Lab 2 Create Ticket > shows field-level validation without calling the API 1060ms
   ✓ Lab 2 Create Ticket > rejects an invalid attachment in the browser 373ms
   ✓ Lab 2 Create Ticket > disables duplicate submission and shows the backend ticket number on success 577ms
   ✓ Lab 2 Create Ticket > keeps entered values after an API failure 313ms
 ✓ tests/lab-04/ActionsTaken.test.tsx (11 tests) 2592ms
   ✓ Lab 4 Actions Taken UI > shows multiple actions in order with performer and assignee clearly separated for a Requester
336ms
   ✓ Lab 4 Actions Taken UI > announces assignee loading and keeps Save disabled until valid choices arrive 406ms
   ✓ Lab 4 Actions Taken UI > lets IT_STAFF create distinct actions with an offset, conditional follow-up and idempotency key
387ms
   ✓ Lab 4 Actions Taken UI > lets ADMINISTRATOR create distinct actions with an offset, conditional follow-up and
idempotency key 358ms
   ✓ Lab 4 Actions Taken UI > keeps form values and focuses invalid input after a server validation error 332ms
 ✓ tests/lab-03/StaffTicketDetail.test.tsx (6 tests) 2524ms
   ✓ Lab 3 Staff Ticket Detail > shows a Requester resolution indication without changing formal status for 'IT_STAFF' 1025ms
   ✓ Lab 3 Staff Ticket Detail > keeps public comments and Internal Notes separate and supports staff operations 797ms
 ✓ tests/lab-04/TicketWorkflow.test.tsx (7 tests) 2752ms
   ✓ Lab 4 Ticket workflow UI > uses only the permitted transitions returned by the server 538ms
   ✓ Lab 4 Ticket workflow UI > shows resolution blockers and links to Actions Taken 670ms
   ✓ Lab 4 Ticket workflow UI > sends the current version, confirmation, and resolution summary 560ms
   ✓ Lab 4 Ticket workflow UI > requires a reopening reason before enabling the status change 314ms


 Test Files  18 passed (18)
      Tests  66 passed (66)
   Start at  13:36:09
   Duration  12.32s (transform 3.45s, setup 26.11s, collect 30.29s, tests 29.53s, environment 99.71s, prepare 5.92s)

Client test exit code: 0

Command: server npm run build; client npm run build; server npx prisma validate

> toktickit-server@1.0.0 build
> tsc

Server build exit code: 0
Prisma schema loaded from prisma\schema.prisma
The schema at prisma\schema.prisma is valid 🚀
Prisma validation exit code: 0

> toktickit-client@1.0.0 build
> tsc && vite build

vite v6.4.3 building for production...
transforming...
```

```
✓ 33 modules transformed.
rendering chunks...
computing gzip size...
dist/index.html                   0.41 kB │ gzip:  0.28 kB
dist/assets/index-CgOHpO3y.css  246.77 kB │ gzip: 34.38 kB
dist/assets/index-Bq3yUVKn.js   238.44 kB │ gzip: 66.41 kB
✓ built in 1.50s
Client build exit code: 0
Validation log complete.
```

End-to-end browser regression

```
Final-main verification: browser regression, 2026-11-01 Asia/Bangkok
Revision: 58b0a7a5556a689ca75c47f9797ebfe84c9cdb3a (merge PR #65)
Tree: 65731368fb2ad8c27e86da34248906a1a5881ee6
Command: npm run test:e2e (fresh disposable local PostgreSQL database)



> toktickit-lab3-integration@1.0.0 test:e2e
> npm run test:e2e:prepare && playwright test



> toktickit-lab3-integration@1.0.0 test:e2e:prepare
> npm --prefix server run test:e2e:prepare



> toktickit-server@1.0.0 test:e2e:prepare
> tsx prisma/prepare-lab3-e2e.ts

New disposable database: toktickit_lab3_e2e_mainlog_20261101133650179_e47151
server: npx --no-install prisma db execute --schema prisma/schema.prisma --stdin
Script executed successfully.
Synthetic pre-Lab-3 snapshot: {"requesters":[{"id":1,"name":"Ariya
Somchai","email":"ariya.somchai@example.com","isActive":true,"createdAt":"2026-11-01T06:36:52.362Z"}],"tickets":
[{"id":1,"ticketNumber":"TT-DOCKER-PRELAB3-
001","requesterId":1,"categoryId":1,"relatedSystemId":1,"requestedPriority":"HIGH","status":"NEW","summary":"Pre-Lab 3
preservation fixture","description":"Synthetic migration fixture for local verification."}],"attachments":
[{"id":1,"ticketId":1,"originalFilename":"baseline.txt","storageKey":"toktickit_lab3_e2e_mainlog_20261101133650179_e47151-
baseline.txt","mimeType":"text/plain","sizeBytes":17,"removedAt":null}],"categories":
[{"id":1,"name":"Hardware","isActive":true,"createdAt":"2026-11-01T06:36:52.362Z"}],"systems":[{"id":1,"name":"Corporate
Laptop","isActive":true,"createdAt":"2026-11-01T06:36:52.362Z"}]}
server: npx --no-install prisma migrate resolve --applied 20260824000000_lab2_foundation
Prisma schema loaded from prisma\schema.prisma
Datasource "db": PostgreSQL database "toktickit_lab3_e2e_mainlog_20261101133650179_e47151", schema "public" at
"localhost:5432"

Migration 20260824000000_lab2_foundation marked as applied.

server: npx --no-install prisma migrate deploy
Prisma schema loaded from prisma\schema.prisma
Datasource "db": PostgreSQL database "toktickit_lab3_e2e_mainlog_20261101133650179_e47151", schema "public" at
"localhost:5432"

4 migrations found in prisma/migrations

Applying migration `20260919000000_lab3_auth_requester`
Applying migration `20261030000000_lab4_actions_foundation`
Applying migration `20261030100000_lab4_ticket_workflow`

The following migration(s) have been applied:

migrations/
  └─ 20260919000000_lab3_auth_requester/
    └─ migration.sql
  └─ 20261030000000_lab4_actions_foundation/
    └─ migration.sql
  └─ 20261030100000_lab4_ticket_workflow/
    └─ migration.sql

All migrations have been successfully applied.
server: npx --no-install prisma migrate status
Prisma schema loaded from prisma\schema.prisma
Datasource "db": PostgreSQL database "toktickit_lab3_e2e_mainlog_20261101133650179_e47151", schema "public" at
"localhost:5432"

4 migrations found in prisma/migrations

Database schema is up to date!
```

```
Post-migration snapshot: {"requesters":[{"id":1,"name":"Ariya
Somchai","email":"ariya.somchai@example.com","isActive":true,"createdAt":"2026-11-01T06:36:52.362Z"}],"tickets":
[{"id":1,"ticketNumber":"TT-DOCKER-PRELAB3-
001","requesterId":1,"categoryId":1,"relatedSystemId":1,"requestedPriority":"HIGH","status":"NEW","summary":"Pre-Lab 3
preservation fixture","description":"Synthetic migration fixture for local verification."}],"attachments":
[{"id":1,"ticketId":1,"originalFilename":"baseline.txt","storageKey":"toktickit_lab3_e2e_mainlog_20261101133650179_e47151-
baseline.txt","mimeType":"text/plain","sizeBytes":17,"removedAt":null}],"categories":
[{"id":1,"name":"Hardware","isActive":true,"createdAt":"2026-11-01T06:36:52.362Z"}],"systems":[{"id":1,"name":"Corporate
Laptop","isActive":true,"createdAt":"2026-11-01T06:36:52.362Z"}]}
server: npm run prisma:seed (run 1)

> toktickit-server@1.0.0 prisma:seed
> tsx prisma/seed.ts

Seeding completed successfully: 10 users and 3 tickets.
Seed 1 assertions:
{"users":10,"tickets":4,"attachments":1,"comments":3,"notes":3,"activeRequesters":4,"inactiveRequesters":1,"activeStaff":3,"i
nactiveStaff":1,"activeAdmins":1,"seededRequesterOwnership":3,"seededStaffOwnership":2,"commentAuthors":3,"noteAuthors":3,"ba
selineAttachment":1}
server: npm run prisma:seed (run 2)

> toktickit-server@1.0.0 prisma:seed
> tsx prisma/seed.ts

Seeding completed successfully: 10 users and 3 tickets.
Seed 2 assertions:
{"users":10,"tickets":4,"attachments":1,"comments":3,"notes":3,"activeRequesters":4,"inactiveRequesters":1,"activeStaff":3,"i
nactiveStaff":1,"activeAdmins":1,"seededRequesterOwnership":3,"seededStaffOwnership":2,"commentAuthors":3,"noteAuthors":3,"ba
selineAttachment":1}
Baseline attachment SHA256: cc7098632c30d31ed81ff5936a75a027f3136024cb96efd35e8fe3f9b4d3cfaf
Prepared E2E fixture IDs/count:
{"databaseName":"toktickit_lab3_e2e_mainlog_20261101133650179_e47151","baselineTicketId":1,"reassignmentOwnerId":8,"queueTota
l":20}
[WebServer] (node:13936) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (Use `node --trace-warnings ...` to show where the warning was created)
[WebServer] (node:27748) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (Use `node --trace-warnings ...` to show where the warning was created)
[WebServer] (node:31508) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (Use `node --trace-warnings ...` to show where the warning was created)

Running 22 tests using 1 worker

(node:18076) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
(Use `node --trace-warnings ...` to show where the warning was created)
  ok  1 [lab-03] › e2e\lab-03\authentication.spec.ts:15:3 › Lab 3 authentication and Requester regression › valid login,
first-login change, role shell, logout, and direct blocking (1.4s)
  ok  2 [lab-03] › e2e\lab-03\authentication.spec.ts:33:3 › Lab 3 authentication and Requester regression › invalid and
inactive accounts receive safe sign-in failure (808ms)
  ok  3 [lab-03] › e2e\lab-03\authentication.spec.ts:50:3 › Lab 3 authentication and Requester regression › Requester can
create, open, comment, and indicate resolution (1.7s)
  ok  4 [lab-03] › e2e\lab-03\authentication.spec.ts:73:3 › Lab 3 authentication and Requester regression › Requester shell
is captured at required responsive sizes (1.2s)
  ok  5 [lab-03] › e2e\lab-03\staff-ticket-flow.spec.ts:5:3 › Lab 3 IT Staff Queue and Ticket Detail › IT Staff can search
Queue, open Detail, and see separate comments and notes (1.7s)
  ok  6 [lab-03] › e2e\lab-03\staff-ticket-flow.spec.ts:23:3 › Lab 3 IT Staff Queue and Ticket Detail › IT Staff Queue and
Detail are captured at required responsive sizes (2.3s)
  ok  7 [lab-03] › e2e\lab-03\staff-ticket-flow.spec.ts:30:3 › Lab 3 IT Staff Queue and Ticket Detail › IT Staff can advance
to a populated second Queue page (1.3s)
  ok  8 [lab-03] › e2e\lab-03\staff-ticket-flow.spec.ts:46:3 › Lab 3 IT Staff Queue and Ticket Detail › preserves migrated
Attachment and verifies reassignment, priority, and status operations (1.5s)
  ok  9 [lab-03] › e2e\lab-03\staff-ticket-flow.spec.ts:75:3 › Lab 3 IT Staff Queue and Ticket Detail › Staff sees the
Requester resolution indication without a formal status change (1.3s)
  ok 10 [lab-03] › e2e\lab-03\user-administration.spec.ts:5:3 › Lab 3 Administrator User Management › Administrator can list,
search, create, edit, reset, and require the new initial password at next login (3.0s)
  ok 11 [lab-03] › e2e\lab-03\user-administration.spec.ts:60:3 › Lab 3 Administrator User Management › Administrator
navigation excludes Queue and direct Ticket Detail remains explicit (735ms)
  ok 12 [lab-03] › e2e\lab-03\user-administration.spec.ts:70:3 › Lab 3 Administrator User Management › Administrator User
Management is captured at required responsive sizes (1.0s)
  ok 13 [lab-03] › e2e\lab-03\z-accessibility-evidence.spec.ts:24:3 › Lab 3 accessibility and transient-state evidence ›
captures the visible sign-in busy state and successful destination (2.0s)
  ok 14 [lab-03] › e2e\lab-03\z-accessibility-evidence.spec.ts:40:3 › Lab 3 accessibility and transient-state evidence ›
completes a keyboard-only sign-in and reaches requester navigation (765ms)
  ok 15 [lab-03] › e2e\lab-03\z-accessibility-evidence.spec.ts:57:3 › Lab 3 accessibility and transient-state evidence ›
traverses the IT Staff Queue and opens Ticket Detail with the keyboard (1.0s)
  ok 16 [lab-03] › e2e\lab-03\z-accessibility-evidence.spec.ts:69:3 › Lab 3 accessibility and transient-state evidence ›
traverses Administrator User Management and opens Create User with the keyboard (994ms)
(node:19384) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
(Use `node --trace-warnings ...` to show where the warning was created)
  ok 17 [lab-02] › e2e\lab-02\requester-ticket-flow.spec.ts:24:3 › Lab 2 requester ticket flow › creates, lists, inspects,
and manages an owned ticket (4.0s)
  ok 18 [lab-02] › e2e\lab-02\requester-ticket-flow.spec.ts:79:3 › Lab 2 requester ticket flow › supports keyboard traversal
across authenticated requester controls (1.3s)
```

```
(node:29320) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
(Use `node --trace-warnings ...` to show where the warning was created)
  ok 19 [lab-04] › e2e\lab-04\actions-taken-flow.spec.ts:27:3 › Lab 4 Actions Taken Ticket Detail › Staff records two actions on one Ticket and its Requester sees both read-only (3.6s)
  ok 20 [lab-04] › e2e\lab-04\dashboards.spec.ts:6:3 › Lab 4 role dashboards › shows Requester and Staff metrics and opens the matching paginated Ticket lists (2.8s)
  ok 21 [lab-04] › e2e\lab-04\ticket-resolution.spec.ts:6:3 › Lab 4 Ticket resolution workflow › resolves, closes, and reopens a Ticket with a Requester-readable history (4.1s)
  ok 22 [lab-04] › e2e\lab-04\zz-accessibility.spec.ts:25:1 › Lab 4 keyboard focus, labels, validation, reduced motion and 320px layouts (3.0s)

  22 passed (49.4s)
E2E exit code: 0
```

Compiled-server health check

```
Final-main verification: compiled server health, 2026-11-01 Asia/Bangkok
Revision: 58b0a7a5556a689ca75c47f9797ebfe84c9cdb3a
Command: node dist/src/index.js; GET /api/health
HTTP status: 200
{"status":"ok","service":"TokTickIT API"}
Health exit code: 0
```